SKTC 1623

Thermodynamics

Chapter 2: Energy, Energy Transfer, and General Energy Analysis

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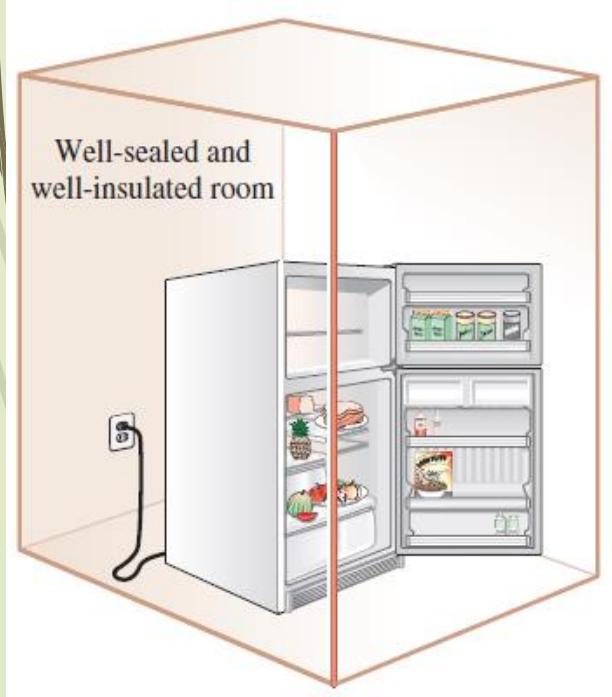
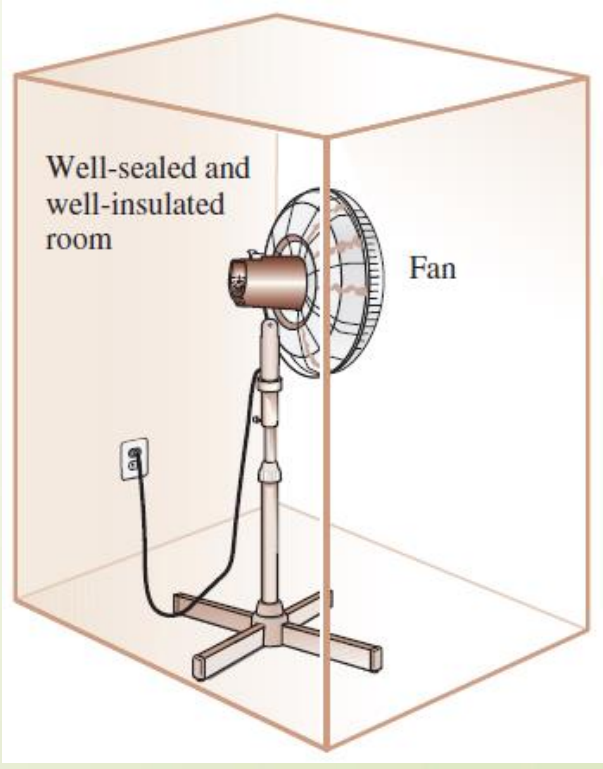
Introduction

- If we take a well-sealed and well-insulated room entirely —including the air and the refrigerator (or fan)—as a system (adiabatic closed system),
  - the only energy interaction involved is the electrical energy crossing the system boundary and entering the room.
- The room temperature will… rise
  - This is as a result of the conversion of electric energy consumed by the device to heat.



A refrigerator operating with its door open in a well-sealed and well-insulated room

A fan running in a well-sealed and well-insulated room will raise the temperature of air in the room.



2.2 Forms of Energy

- **Energy can exist in numerous forms** such as thermal, mechanical, kinetic, potential, electric, magnetic, chemical, and nuclear, and their sum constituents the **total energy, E** of a system.
- **Thermodynamics** provides no information about the absolute value of the total energy. It deals only with **the change of the total energy**.
- In thermodynamic analysis, the various forms of energy that make up the total energy of a system is considered in two groups: *macroscopic* and *microscopic.*
- **Macroscopic** forms of energy are those a system possesses as a whole with respect to some outside reference frame, such as **kinetic** and **potential** energies.
- **Microscopic** forms of energy are those related to the molecular structure of a system and the degree of the molecular activity, and they are independent of outside reference frames.

2.2 Forms of Energy (2)

- The sum of all the microscopic forms of energy is called the **internal energy** of a system and is denoted by *U.*
- **Kinetic energy, KE**: The energy that a system possesses as a result of its motion relative to some reference frame.
- **Potential energy, PE**: The energy that a system possesses as a result of its elevation in a gravitational field.

2.2 Forms of Energy (3)

Kinetic energy

- When all parts of a system move with the same velocity, the kinetic energy is expressed as $\mathrm{KE} = m\dfrac{V^2}{2} \qquad (\mathrm{kJ})$

  or, on a unit mass basis, $\mathrm{ke} = \dfrac{V^2}{2} \qquad (\mathrm{kJ/kg})$

  where V denotes the velocity of the system relative to some fixed reference frame.

Potential energy

- is expressed as $\mathrm{PE} = mgz \qquad (\mathrm{kJ})$
- or, on a unit mass basis, $\mathrm{pe} = gz \qquad (\mathrm{kJ/kg})$

  where g is the gravitational acceleration and z is the elevation of the center of gravity of a system relative to some arbitrarily selected reference level.

2.2 Forms of Energy (4)

Total energy

- The total energy of a common system consists of the kinetic, potential, and internal energies and is expressed as

$$E = U + \mathrm{KE} + \mathrm{PE} = U + m\frac{V^2}{2} + mgz \qquad (\mathrm{kJ})$$

  - The other forms of energy usually are neglected, unless specified in the problem.
- The total energy of a system on a unit mass basis is denoted by e and is defined as:

$$e = \frac{E}{m} \qquad (\mathrm{kJ/kg})$$

- So the total energy per unit mass:

$$e = u + \mathrm{ke} + \mathrm{pe} = u + \frac{V^2}{2} + gz \qquad (\mathrm{kJ/kg})$$

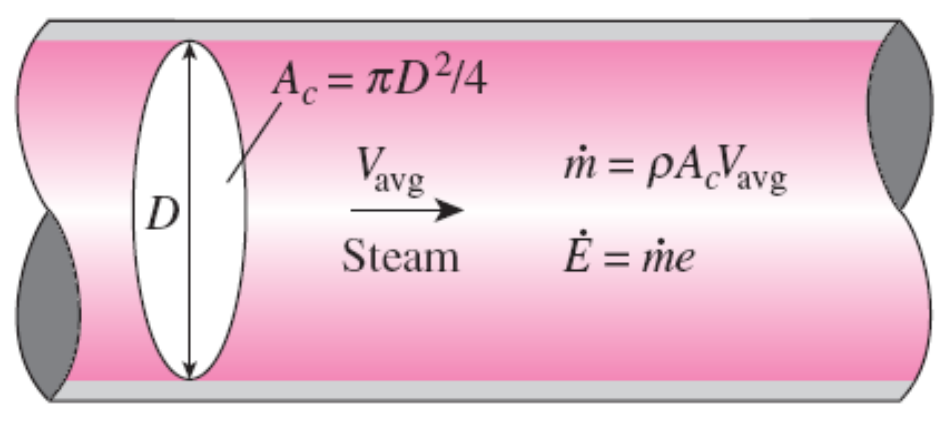
2.2 Forms of Energy (5)

- A **closed systems** whose velocity and elevation of the center of gravity remain constant during a process is referred to as **Stationary systems .**
  - The change in the total energy ΔE is identical to the change in its internal energy ΔU since there is no change in the kinetic and potential energies.
- **Control volumes** typically involve fluid flow for long periods of time, so better to express **energy flow** associated with a fluid stream in the rate form:
  - **Mass flow rate,** $\dot{m}$ is the amount of mass flowing through a cross section per unit time.

$$\dot{m} = \rho \dot{V} = \rho A_c V_{avg} \qquad (\text{kg/s})$$

  - **Volume flow rate,** $\dot{V}$ is the volume of the fluid flowing through a cross section per unit time.
  - **Energy flow rate:**

$$\dot{E} = \dot{m}e \qquad (\text{kJ/s or kW})$$



2.2 Forms of Energy (6)

Some Physical Insight to Internal Energy

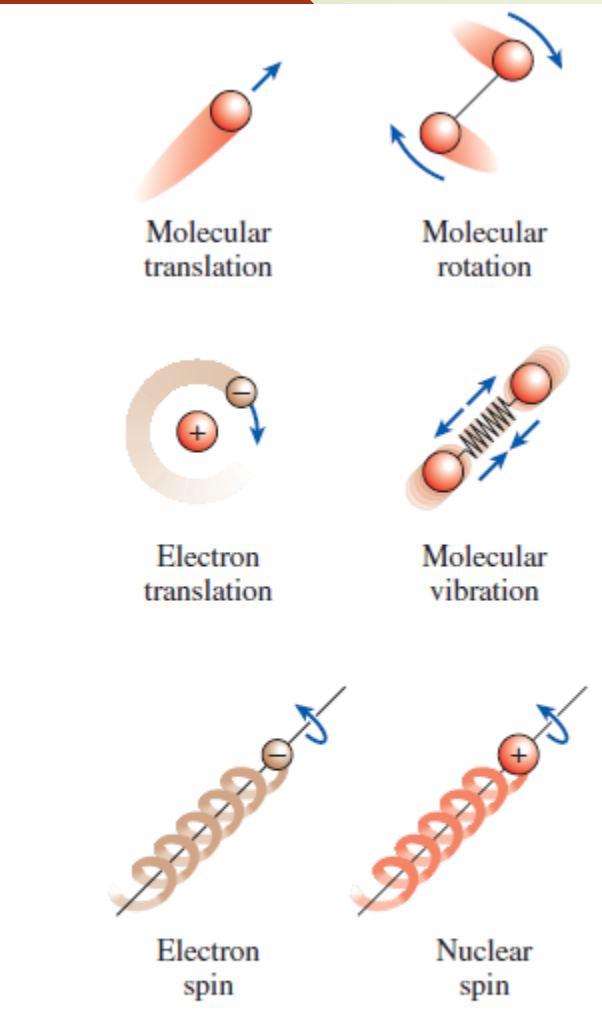


FIGURE 2–6
The various forms of microscopic energies that make up *sensible* energy.

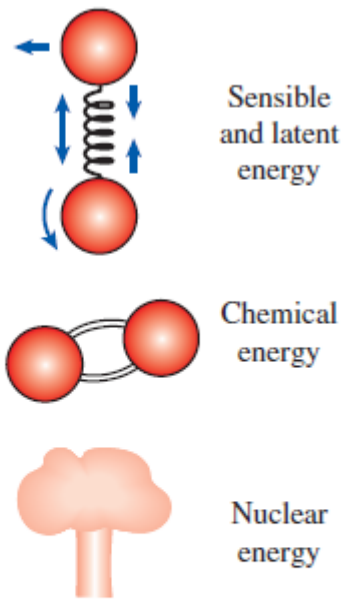


FIGURE 2–7
The internal energy of a system is the sum of all forms of the microscopic energies.

- **Sensible energy:** The portion of the internal energy of a system associated with the kinetic energies of the molecules.
- **Latent energy:** The internal energy associated with the phase of a system.
- **Chemical energy:** The internal energy associated with the atomic bonds in a molecule.
- **Nuclear energy:** The tremendous amount of energy associated with the strong bonds within the nucleus of the atom itself.

Thermal = Sensible + Latent

Internal = Sensible + Latent + Chemical + Nuclear

2.2 Forms of Energy (7)

- The total energy of a system, can be *contained* or *stored* in a system, and thus can be viewed as the ***static* forms of energy**.
- The forms of energy not stored in a system can be viewed as the ***dynamic* forms of energy** or as ***energy interactions***.
- The dynamic forms of energy are recognized at the system boundary as they cross it, and represent the energy gained or lost by a system during a process.
- The only two forms of energy interactions associated with a closed system are **heat transfer** and **work**.
  - An energy interaction is heat transfer if its driving force is a temperature difference.
  - Otherwise it is work.

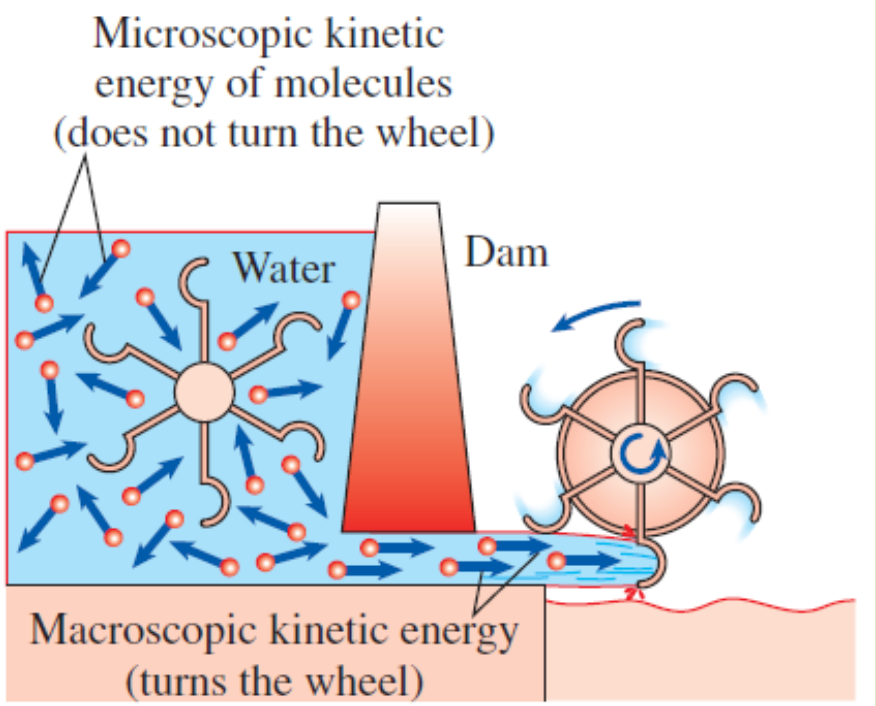
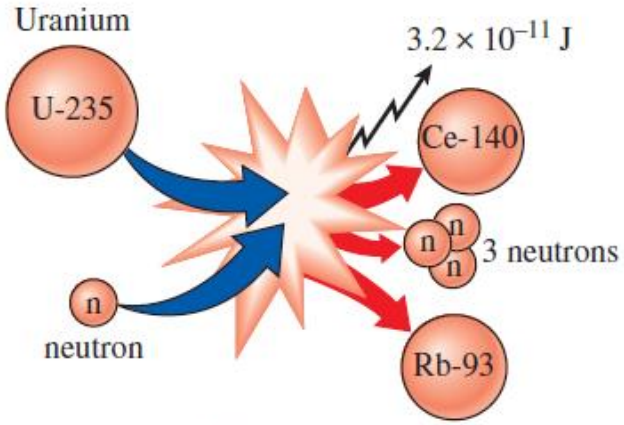


FIGURE 2–8
The *macroscopic* kinetic energy is an organized form of energy and is much more useful than the disorganized *microscopic* kinetic energies of the molecules.

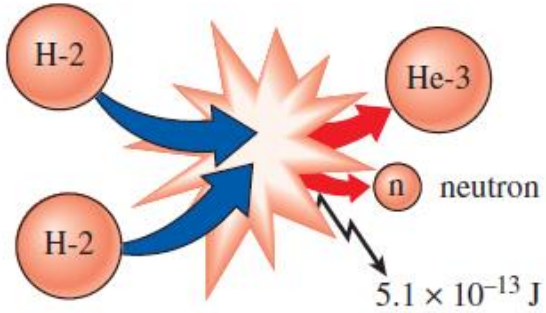
2.2 Forms of Energy (8)

More on Nuclear Energy

- The best known **fission** reaction involves the split of the uranium atom (the U-235 isotope) into other elements
  - The complete fission of 1 kg of U-235 releases 6.73 x 10^{10} kJ of heat (more than heat released from 3000 tons of coal).
  - Commonly used to generate electricity in nuclear power plants, to power nuclear submarines and aircraft carriers, and even to power spacecraft as well as building nuclear bombs.
- Nuclear energy by **fusion** is released when two small nuclei combine into a larger one.
  - The uncontrolled fusion reaction was achieved in the early 1950s, but all the efforts since then to achieve controlled fusion by massive lasers, powerful magnetic fields, and electric currents to generate power have failed.



(*a*) Fission of uranium



(*b*) Fusion of hydrogen

FIGURE 2–9
The fission of uranium and the fusion of hydrogen during nuclear reactions, and the release of nuclear energy.

EXAMPLE 2–1 A Car Powered by Nuclear Fuel

An average car consumes about 5 L of gasoline a day, and the capacity of the fuel tank of a car is about 50 L. Therefore, a car needs to be refueled once every 10 days. Also, the density of gasoline ranges from 0.68 to 0.78 kg/L, and its lower heating value is about 44,000 kJ/kg (that is, 44,000 kJ of heat is released when 1 kg of gasoline is completely burned). Suppose all the problems associated with the radioactivity and waste disposal of nuclear fuels are resolved, and a car is to be powered by U-235. If a new car comes equipped with 0.1-kg of the nuclear fuel U-235, determine if this car will ever need refueling under average driving conditions (Fig. 2–10).

FIGURE 2–10
Schematic for Example 2–1.

SOLUTION A car powered by nuclear energy comes equipped with nuclear fuel. It is to be determined if this car will ever need refueling.

Assumptions **1** Gasoline is an incompressible substance with an average density of 0.75 kg/L. **2** Nuclear fuel is completely converted to thermal energy.

Analysis The mass of gasoline used per day by the car is

$$m_{\text{gasoline}} = (\rho V)_{\text{gasoline}} = (0.75\ \text{kg/L})(5\ \text{L/day}) = 3.75\ \text{kg/day}$$

Noting that the heating value of gasoline is 44,000 kJ/kg, the energy supplied to the car per day is

$$E = (m_{\text{gasoline}})(\text{Heating value})$$
$$= (3.75\ \text{kg/day})(44{,}000\ \text{kJ/kg}) = 165{,}000\ \text{kJ/day}$$

The complete fission of 0.1 kg of uranium-235 releases

$$(6.73 \times 10^{10}\ \text{kJ/kg})(0.1\ \text{kg}) = 6.73 \times 10^{9}\ \text{kJ}$$

of heat, which is sufficient to meet the energy needs of the car for

$$\text{No. of days} = \frac{\text{Energy content of fuel}}{\text{Daily energy use}} = \frac{6.73 \times 10^{9}\ \text{kJ}}{165{,}000\ \text{kJ/day}} = \mathbf{40{,}790\ days}$$

which is equivalent to about 112 years. Considering that no car will last more than 100 years, this car will never need refueling. It appears that nuclear fuel of the size of a cherry is sufficient to power a car during its lifetime.

Discussion Note that this problem is not quite realistic since the necessary critical mass cannot be achieved with such a small amount of fuel. Further, all of the uranium cannot be converted in fission, again because of the critical mass problems after partial conversion.

FIGURE 2–10
Schematic for Example 2–1.

2.2 Forms of Energy (9)

Mechanical Energy

- **Mechanical energy** is the **form of energy that can be converted to mechanical work completely and directly** by an ideal mechanical device such as an ideal turbine.
- Kinetic and potential energies are the familiar forms of mechanical energy.
- Thermal energy is not mechanical energy (cannot be converted to work directly and completely)
- Pump – transfer mechanical energy to a fluid by raising its pressure
- Turbine – extracts mechanical energy from fluid by dropping its pressure
- Pressure itself is not a form of energy.
  - But a pressure force acting on a fluid through a distance produces work, called flow work, in the amount of P/ρ per unit mass.

2.2 Forms of Energy (10)

- The mechanical energy of a flowing fluid can be expressed on a unit mass basis as :

$$e_{mech} = \frac{P}{\rho} + \frac{V^2}{2} + gz$$

where:
P/ρ is the flow energy,
$V^2/2$ is the kinetic energy, and
gz is the potential energy of the fluid, all per unit mass.

in rate form,

$$\dot{E}_{\text{mech}} = \dot{m}e_{\text{mech}} = \dot{m}\left(\frac{P}{\rho} + \frac{V^2}{2} + gz\right)$$

where $\dot{m}$ is the mass flow rate of the fluid.

- The mechanical energy change of a fluid during incompressible (ρ = constant) flow becomes:

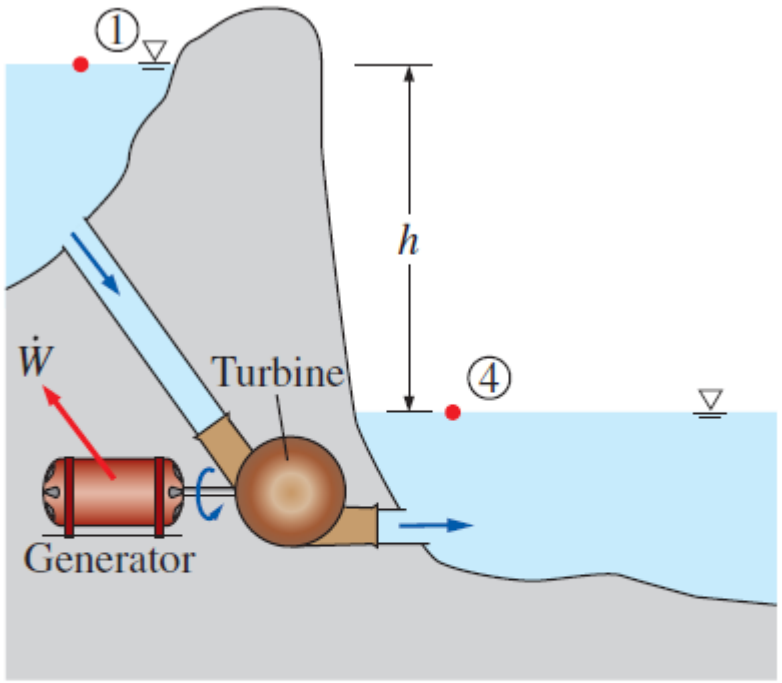
$$\Delta e_{\text{mech}} = \frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \qquad (\text{kJ/kg})$$

in rate form,

$$\Delta \dot{E}_{\text{mech}} = \dot{m}\Delta e_{\text{mech}} = \dot{m}\left(\frac{P_2 - P_1}{\rho} + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)\right) \qquad (\text{kW})$$

- Mechanical energy of a fluid does not change during flow if its pressure, density, velocity and elevation remain constant.

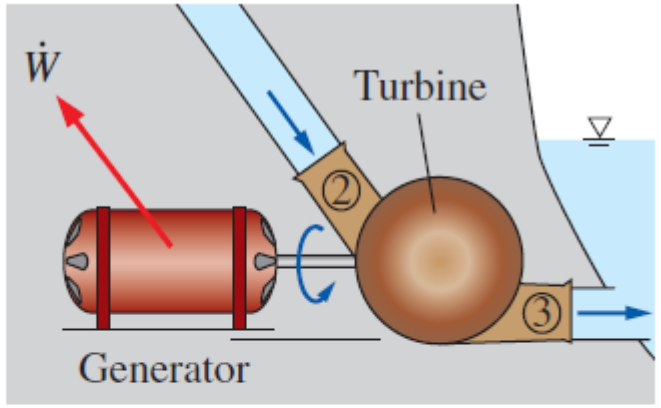
2.2 Forms of Energy (11)



$$\dot{W}_{max} = \dot{m}\Delta e_{mech} = \dot{m}g(z_1 - z_4) = \dot{m}gh$$

since $P_1 \approx P_4 = P_{atm}$ and $V_1 = V_4 \approx 0$

(*a*)



$$\dot{W}_{max} = \dot{m}\Delta e_{mech} = \dot{m}\frac{P_2 - P_3}{\rho} = \dot{m}\frac{\Delta P}{\rho}$$

since $V_2 \approx V_3$ and $z_2 = z_3$

(*b*)

FIGURE 2–12

Mechanical energy is illustrated by an ideal hydraulic turbine coupled with an ideal generator. In the absence of irreversible losses, the maximum produced power is proportional to (*a*) the change in water surface elevation from the upstream to the downstream reservoir or (*b*) (close-up view) the drop in water pressure from just upstream to just downstream of the turbine.

EXAMPLE 2–2 Wind Energy

A site evaluated for a wind farm is observed to have steady winds at a speed of 8.5 m/s (Fig. 2–13). Determine the wind energy (*a*) per unit mass, (*b*) for a mass of 10 kg, and (*c*) for a flow rate of 1154 kg/s for air.

SOLUTION A site with a specified wind speed is considered. Wind energy per unit mass, for a specified mass, and for a given mass flow rate of air are to be determined.

Assumptions Wind flows steadily at the specified speed.

Analysis The only harvestable form of energy of atmospheric air is the kinetic energy, which is captured by a wind turbine.

(*a*) Wind energy per unit mass of air is

$$e = \text{ke} = \frac{V^2}{2} = \frac{(8.5\ \text{m/s})^2}{2}\left(\frac{1\ \text{J/kg}}{1\ \text{m}^2/\text{s}^2}\right) = \mathbf{36.1\ J/kg}$$

(*b*) Wind energy for an air mass of 10 kg is

$$E = me = (10\ \text{kg})(36.1\ \text{J/kg}) = \mathbf{361\ J}$$

(*c*) Wind energy for a mass flow rate of 1154 kg/s is

$$\dot{E} = \dot{m}e = (1154\ \text{kg/s})(36.1\ \text{J/kg})\left(\frac{1\ \text{kW}}{1000\ \text{J/s}}\right) = \mathbf{41.7\ kW}$$

FIGURE 2–13
A site for a wind farm as discussed in Example 2–2.

Discussion It can be shown that the specified mass flow rate corresponds to a 12-m diameter flow section when the air density is 1.2 kg/m³. Therefore, a wind turbine with a wind span diameter of 12 m has a power generation potential of 41.7 kW. Real wind turbines convert about one-third of this potential to electric power.

2.3 Energy Transfer by Heat

- **Heat** is defined as the form of energy that is transferred between two systems (or a system and its surroundings) by virtue of a temperature difference.

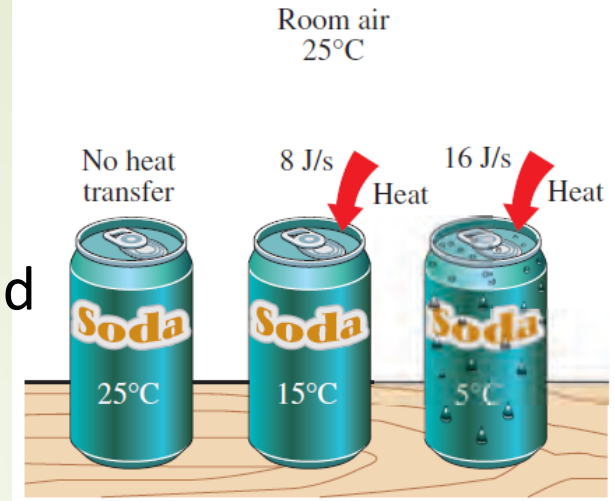


FIGURE 2–15
Temperature difference is the driving force for heat transfer. The larger the temperature difference, the higher is the rate of heat transfer.

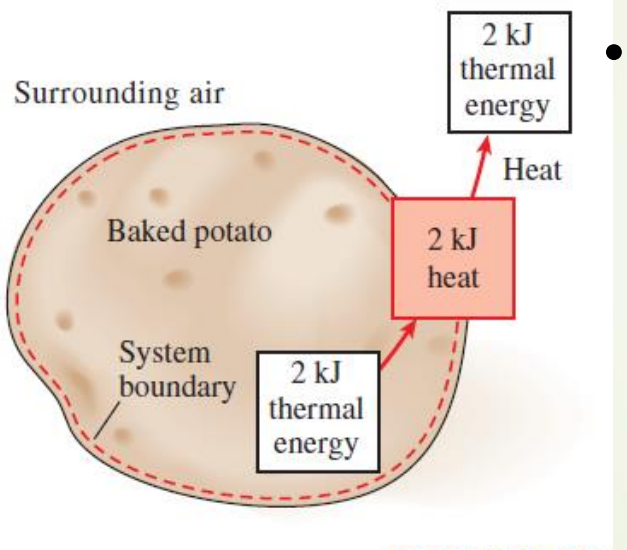


FIGURE 2–16
Energy is recognized as heat transfer only as it crosses the system boundary.

- Heat is energy in transition
  - Only recognized when it crosses the boundary of a system
  - **Adiabatic process** is a process during which there is no heat transfer.
    - The system is well insulated, or
    - Both system and its surroundings are at the same temperature

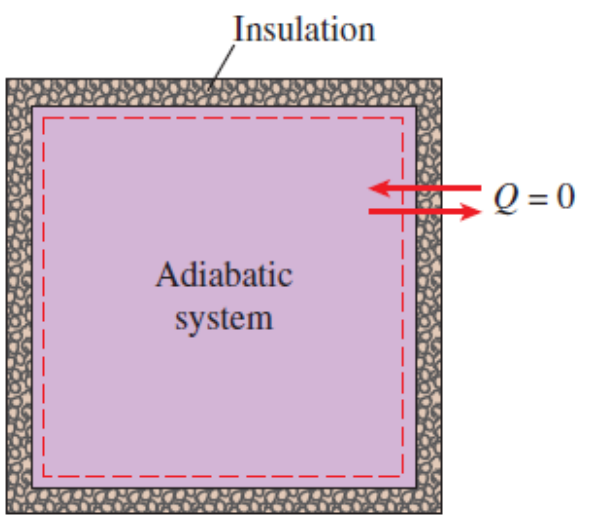


FIGURE 2–17
During an adiabatic process, a system exchanges no heat with its surroundings.

2.3 Energy Transfer by Heat (2)

- Heat transfer per unit mass, $q = \dfrac{Q}{m} \qquad (\text{kJ/kg})$
- Amount of heat transfer when heat transfer rate is constant

$$Q = \dot{Q}\ \Delta t \qquad (\text{kJ})$$

- Amount of heat transfer when heat transfer rate changes with time

$$Q = \int_{t_1}^{t_2} \dot{Q}\, dt \qquad (\text{kJ})$$

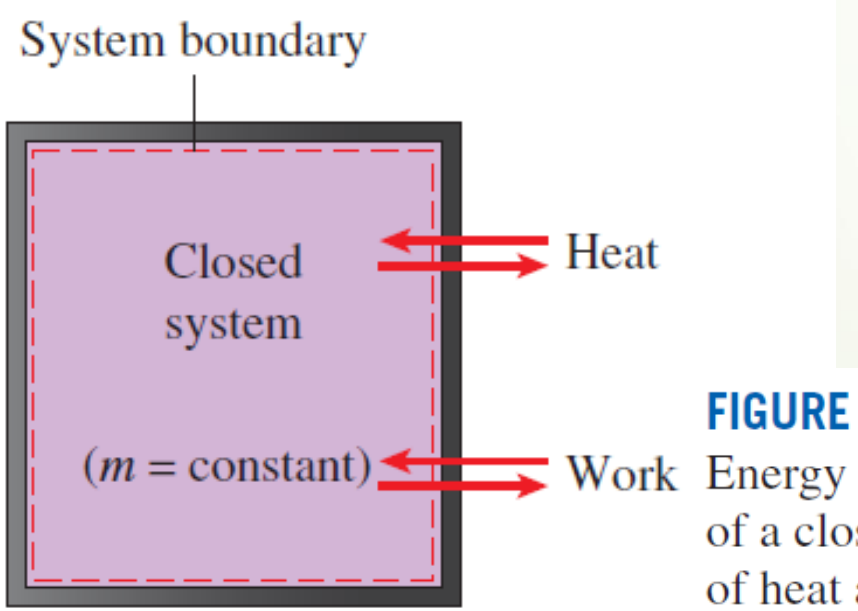


FIGURE 2–14
Energy can cross the boundaries of a closed system in the form of heat and work.

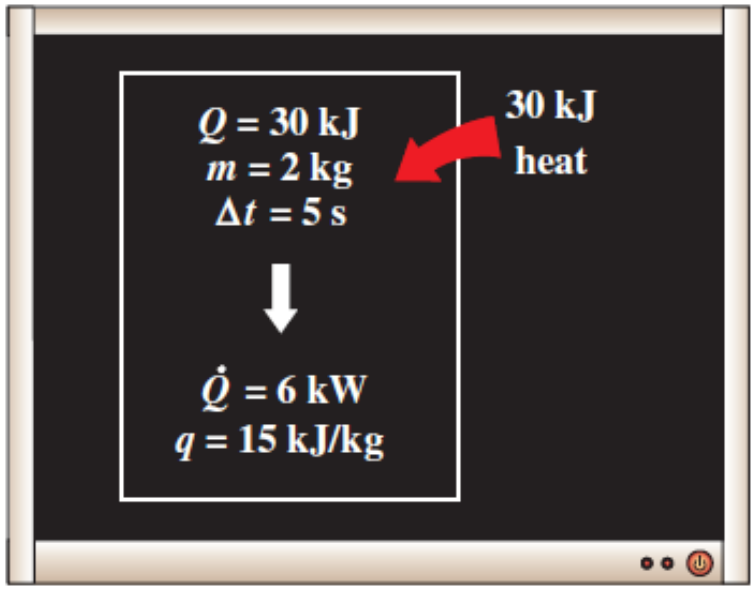


FIGURE 2–18
The relationships among q, Q, and $\dot{Q}$.

2.3 Energy Transfer by Heat (3)

Heat transfer mechanisms:

- **Conduction**: The transfer of energy from the more energetic particles of a substance to the adjacent less energetic ones as a result of interaction between particles.
- **Convection**: The transfer of energy between a solid surface and the adjacent fluid that is in motion, and it involves the combined effects of conduction and fluid motion.
- **Radiation**: The transfer of energy due to the emission of electromagnetic waves (or photons).

2.4 Energy Transfer by Work

- **Work**, like **heat** , is an **energy interaction** between a system and its surroundings.
- Work is the energy transfer associated with a force acting through a distance.
  - A rising piston, a rotating shaft, and an electric wire crossing the system boundaries are all associated with work interactions
- The work done during a process between states 1 and 2 is denoted by W_{12}, or simply W.
- The work done per unit mass of a system is denoted by w and is expressed as $w = W/m$ (kJ/kg).

$$w = \frac{W}{m} \qquad (\text{kJ/kg})$$

- The work done per unit time is called **power** and is denoted by $\dot{W}$ (kJ/s or kW)

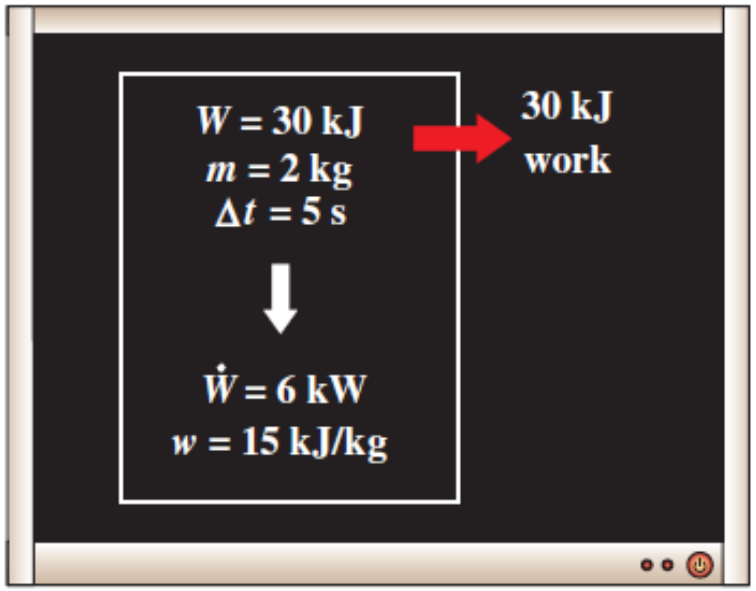


FIGURE 2–20
The relationships among w, W, and $\dot{W}$.

2.4 Energy Transfer by Work (2)

- Formal **sign convention** (classical thermodynamics sign convention):
  - heat transfer to a system and work done by a system are positive;
  - heat transfer from a system and work done on a system are negative.
- Alternative to sign convention is to use the subscripts in and out to indicate direction.
  - This is the primary approach in this text.

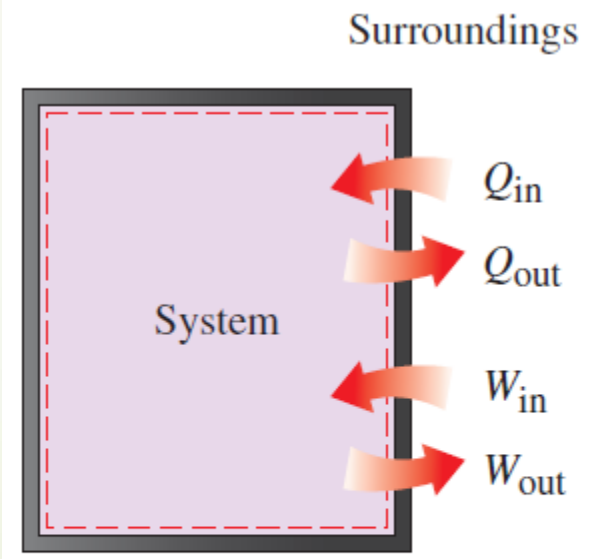


FIGURE 2–21
Specifying the directions of heat and work.

2.4 Energy Transfer by Work (3)

- Heat and work are ***energy transfer mechanisms*** between a system and its surroundings, and there are many similarities between them:
  - Both are recognized at the boundaries of a system as they cross the boundaries. That is, both heat and work are *boundary* phenomena.
  - Systems possess energy, but not heat or work.
  - Both are associated with a *process*, not a state.
    - Unlike properties, heat or work has no meaning at a state.
  - Both are *path functions* (i.e., their magnitudes depend on the path followed during a process as well as the end states).
- **Path functions** have **inexact differentials** designated by the symbol δ.
  - Therefore, a differential amount of heat or work is represented by δQ or δW, respectively, instead of dQ or dW.

2.4 Energy Transfer by Work (4)

- Properties, however, are **point functions** (they depend on the state only), and they have **exact differentials** designated by the symbol *d*.
- A small change in volume, is represented by *dV,* and the total volume change during a process between states 1 and 2 is

$$\int_1^2 dV = V_2 - V_1 = \Delta V$$

- The volume change is always the volume at state 2 minus volume at state 1, regardless of the path followed.
- The total work done during process 1-2, is

$$\int_1^2 \delta W = W_{12} \qquad (not\, \Delta W)$$

- The total work is obtained by following the process path and adding the differential amounts of work *(δW)* done along the way. **The integral of *δW is not* $W_2 - W_1$.**

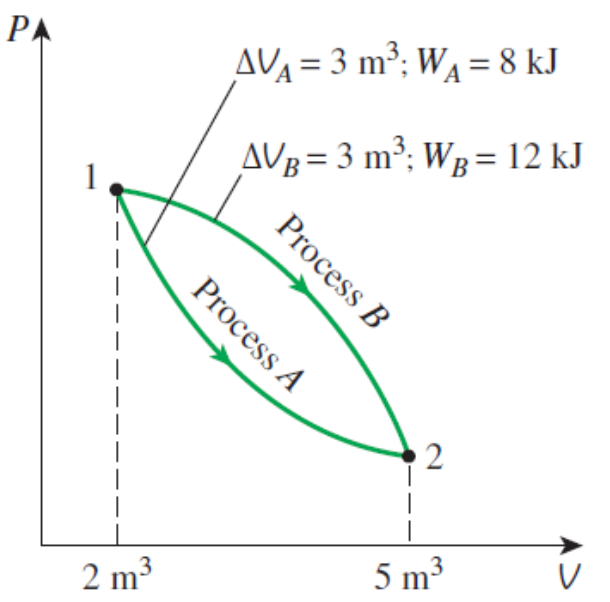


FIGURE 2–22
Properties are point functions; but heat and work are path functions (their magnitudes depend on the path followed).

2.4 Energy Transfer by Work (5)

Electrical Work

- In an electric field, electrons in a wire move under the effect of electromotive forces, doing work
- When *N* coulombs of electrical charge move through a potential difference *V*, the electrical work done is:

$$W_e = \mathbf{V}N \text{ (kJ) (Electrical work)}$$

- Which can also be expressed in the rate form as:

$$\dot{W}_e = \mathbf{V}I \qquad (\mathrm{W}) \text{ (Electrical power)}$$

where *I* is the number of electrical charges flowing per unit time (current)

- When potential difference and current change with time

$$W_e = \int_1^2 \mathbf{V}I\,dt \qquad (\mathrm{kJ})$$

- When potential difference and current remain constant

$$W_e = \mathbf{V}I\,\Delta t \qquad (\mathrm{kJ})$$

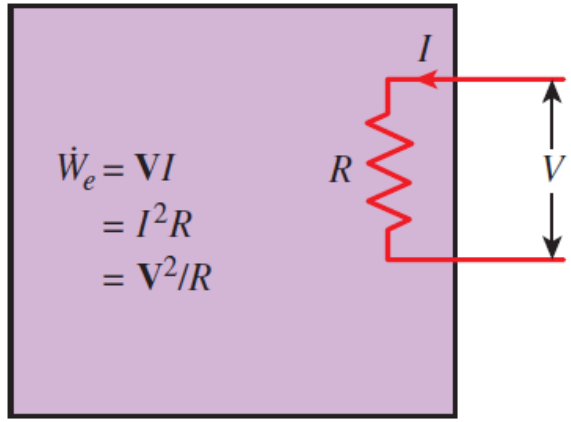


FIGURE 2–27
Electrical power in terms of resistance *R*, current *I*, and potential difference **V**.

2.5 Mechanical Form of Work

- There are two requirements for a work interaction between a system and its surroundings to exist:
  - there must be a ***force*** acting on the boundary.
  - the boundary must ***move***.

Work = Force × Distance

$$W = Fs \qquad (\text{kJ})$$

When force is not constant

$$W = \int_1^2 F\, ds \qquad (\text{kJ})$$

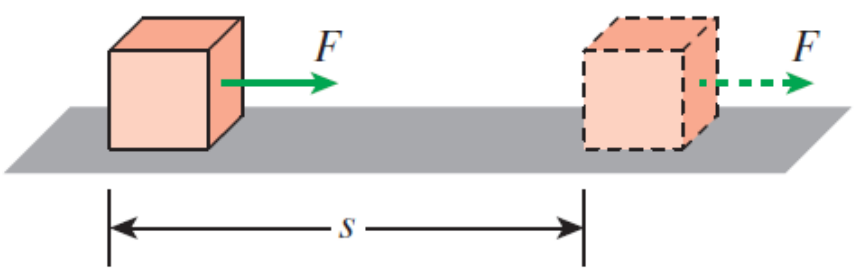


FIGURE 2–28
The work done is proportional to the force applied (F) and the distance traveled (s).

2.5 Mechanical Form of Work (2)

Shaft Work

- Energy transmission with a rotating shaft is very common in engineering practice
- A force *F* acting through a moment arm *r* generates a torque T

$$\mathrm{T} = Fr \quad \rightarrow \quad F = \frac{\mathrm{T}}{r}$$

- This force acts through a distance, $s = (2\pi r)n$
- Then the shaft work is

$$W_{\mathrm{sh}} = Fs = \left(\frac{\mathrm{T}}{r}\right)(2\pi rn) = 2\pi n\mathrm{T} \qquad (\mathrm{kJ})$$

- The power transmitted through the shaft is the shaft work done per unit time

$$\dot{W}_{\mathrm{sh}} = 2\pi \dot{n}\mathrm{T} \qquad (\mathrm{kW})$$

where $\dot{n}$ is the number of revolutions per unit time.

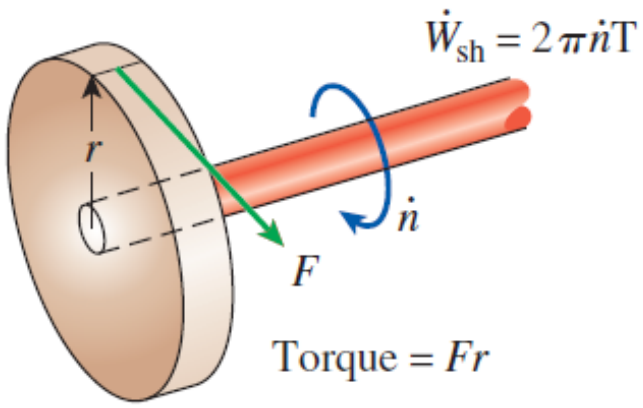


FIGURE 2–30
Shaft work is proportional to the torque applied and the number of revolutions of the shaft.

EXAMPLE 2–7 Power Transmission by the Shaft of a Car

Determine the power transmitted through the shaft of a car when the torque applied is 200 N·m and the shaft rotates at a rate of 4000 revolutions per minute (rpm).

SOLUTION The torque and the rpm for a car engine are given. The power transmitted is to be determined.

Analysis A sketch of the car is given in Fig. 2–31. The shaft power is determined directly from

$$\dot{W}_{sh} = 2\pi\dot{n}T = (2\pi)\left(4000\ \frac{1}{\text{min}}\right)(200\ \text{N·m})\left(\frac{1\ \text{min}}{60\ \text{s}}\right)\left(\frac{1\ \text{kJ}}{1000\ \text{N·m}}\right)$$

$$= \mathbf{83.8\ kW} \quad (\text{or } 112\ \text{hp})$$

Discussion Note that power transmitted by a shaft is proportional to torque and the rotational speed.

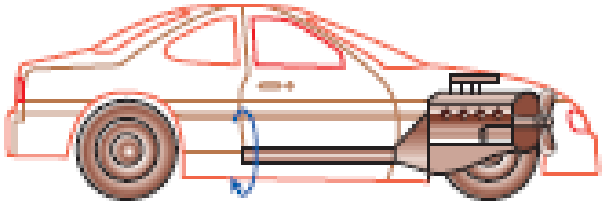


FIGURE 2–31
Schematic for Example 2–7.

2.5 Mechanical Form of Work (3)

Spring Work

- When the length of the spring changes by a differential amount *dx* under the influence of a force *F*, the work done is $\delta W_{\text{spring}} = F\,dx$
- For linear elastic springs, the displacement *x* is proportional to the force applied

$$F = kx \qquad (\text{kN})$$

k: spring constant (kN/m)

- Substituting and integrating yield

$$W_{\text{spring}} = \tfrac{1}{2}k(x_2^2 - x_1^2) \qquad (\text{kJ})$$

where x_1 and x_2: the initial and the final displacements

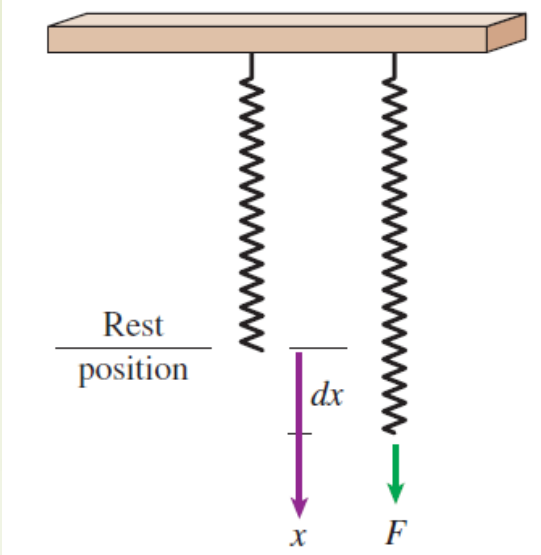


FIGURE 2–32
Elongation of a spring under the influence of a force.

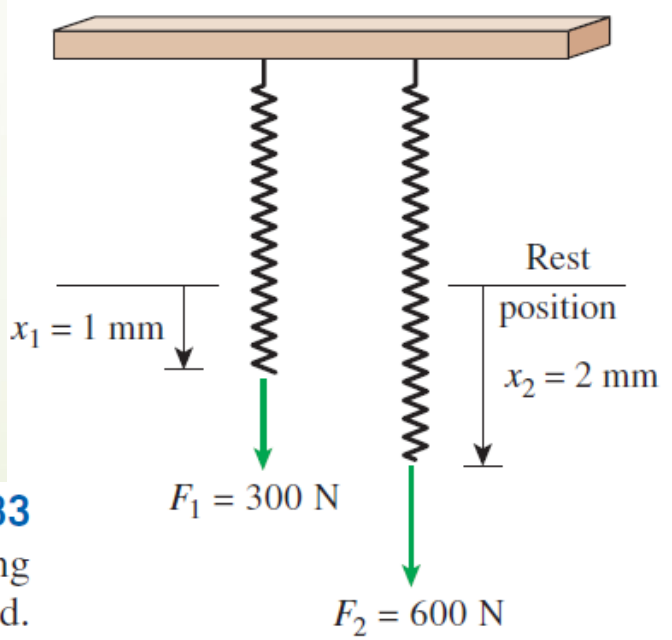


FIGURE 2–33
The displacement of a linear spring doubles when the force is doubled.

2.5 Mechanical Form of Work (4)

Work Done to Raise or to Accelerate a Body

- The conservation of energy principle requires that an equivalent amount of energy must be transferred to the body being raised or accelerated.
  - The work transfer needed to raise a body is equal to the change in the potential energy of the body.
  - The work transfer needed to accelerate a body is equal to the change in the kinetic energy of the body.
  - Similarly, the potential or kinetic energy of a body represents the work that can be obtained from the body as it is lowered to the reference level or decelerated to zero velocity.

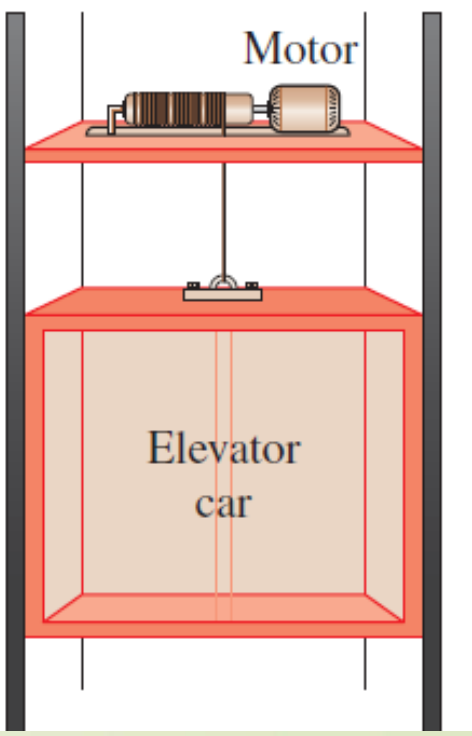


FIGURE 2–36
The energy transferred to a body while being raised is equal to the change in its potential energy.

EXAMPLE 2–8 Power Needs of a Car to Climb a Hill

A man whose mass is 100 kg pushes a cart whose mass, including its contents, is 100 kg up a ramp that is inclined at an angle of 20° from the horizontal (Fig. 2–37). The local gravitational acceleration is 9.8 m/s^2. Determine the work, in kJ, needed to move along this ramp a distance of 100 m considering (*a*) the man and (*b*) the cart and its contents as the system.

SOLUTION A man is pushing a cart with its contents up a ramp that is inclined at an angle of 20° from the horizontal. The work needed to move along this ramp is to be determined considering (*a*) the man and (*b*) the cart and its contents as the system.

Analysis (*a*) Considering the man as the system, letting l be the displacement along the ramp, and letting θ be the inclination angle of the ramp,

$$W = Fl\sin\theta = mgl\sin\theta$$
$$= (100 + 100\text{ kg})(9.8\text{ m/s}^2)(100\text{ m})(\sin 20°)\left(\frac{1\text{ kJ/kg}}{1000\text{ m}^2/\text{s}^2}\right) = \mathbf{67.0\text{ kJ}}$$

This is work that the man must do to raise the weight of the cart and contents, plus his own weight, a distance of $l\sin\theta$.

(*b*) Applying the same logic to the cart and its contents gives

$$W = Fl\sin\theta = mgl\sin\theta$$
$$= (100\text{ kg})(9.8\text{ m/s}^2)(100\text{ m})(\sin 20°)\left(\frac{1\text{ kJ/kg}}{1000\text{ m}^2/\text{s}^2}\right) = \mathbf{33.5\text{ kJ}}$$

Discussion The result in part (*a*) is more realistic since the man has to move himself in addition to the cart.

FIGURE 2–37
Schematic for Example 2–8.

EXAMPLE 2–9 Power Needs of a Car to Accelerate

Determine the power required to accelerate a 900-kg car shown in Fig. 2–38 from rest to a velocity of 80 km/h in 20 s on a level road.

SOLUTION The power required to accelerate a car to a specified velocity is to be determined.

Analysis The work needed to accelerate a body is simply the change in the kinetic energy of the body,

$$W_a = \tfrac{1}{2}m(V_2^2 - V_1^2) = \tfrac{1}{2}(900\text{ kg})\left[\left(\frac{80{,}000\text{ m}}{3600\text{ s}}\right)^2 - 0^2\right]\left(\frac{1\text{ kJ/kg}}{1000\text{ m}^2/\text{s}^2}\right)$$

$$= 222\text{ kJ}$$

The average power is determined from

$$\dot{W}_a = \frac{W_a}{\Delta t} = \frac{222\text{ kJ}}{20\text{ s}} = \mathbf{11.1\ kW} \qquad (\text{or } 14.9\text{ hp})$$

Discussion This is in addition to the power required to overcome friction, rolling resistance, and other imperfections.

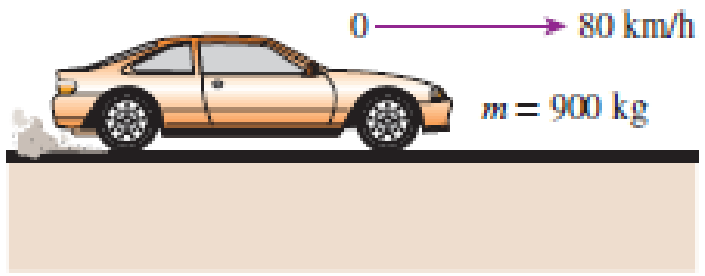


FIGURE 2–38
Schematic for Example 2–9.

2.6 The First law of Thermodynamics

- **First law of thermodynamics** (or the conservation of energy principle) states that:

  Energy can be neither created nor destroyed during a process; it can only change forms.

- For all **adiabatic processes** between two specified states of a closed system, the **net work done** is the same regardless of the nature of the closed system and the details of the process.

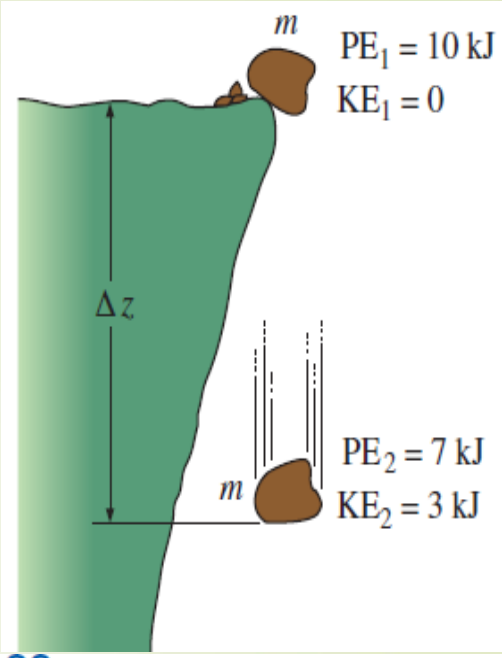


FIGURE 2–39
Energy cannot be created or destroyed; it can only change forms.

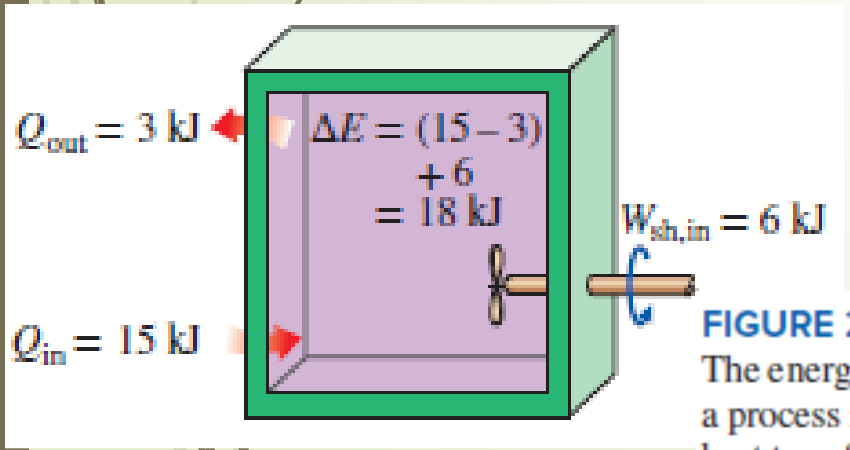


FIGURE 2–45
The energy change of a system during a process is equal to the *net* work and heat transfer between the system and its surroundings.

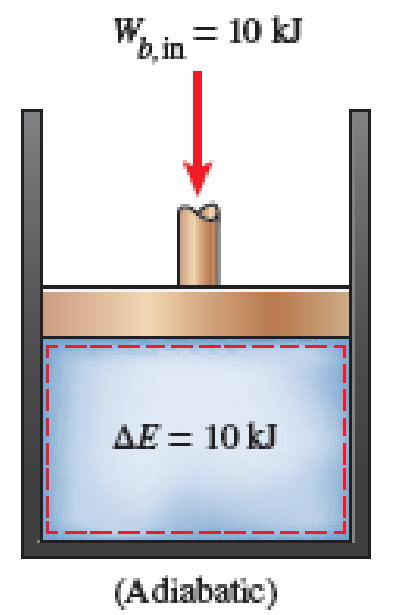


FIGURE 2–44
The work (boundary) done on an adiabatic system is equal to the increase in the energy of the system.

2.6 The First law of Thermodynamics (2)

Energy Balance

- The net change (increase or decrease) in the total energy of the system during a process is equal to the difference between the total energy entering and the total energy leaving the system during that process.

$$\begin{pmatrix}\text{Total energy}\\ \text{entering the system}\end{pmatrix} - \begin{pmatrix}\text{Total energy}\\ \text{leaving the system}\end{pmatrix} = \begin{pmatrix}\text{Change in the total}\\ \text{energy of the system}\end{pmatrix}$$

$$E_{in} - E_{out} = \Delta E_{system}$$

$$\boxed{E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out}) + (E_{mass,in} - E_{mass,out}) = \Delta E_{system}}$$

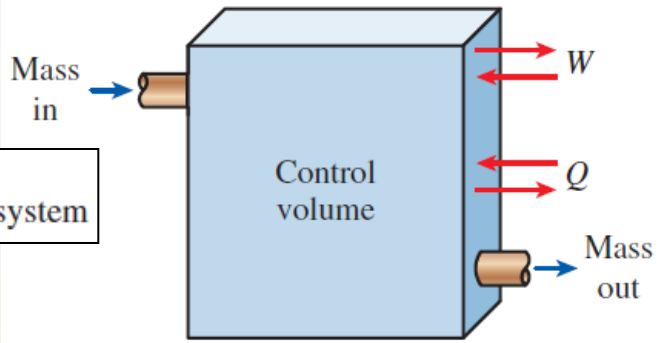


FIGURE 2–47
The energy content of a control volume can be changed by mass flow as well as heat and work interactions.

- The successful use of this relation to solve engineering problems depends on understanding the various forms of energy and recognizing the forms of energy transfer.

2.6 The First law of Thermodynamics (3)

Energy Change of a System, ΔE_{system}

- The determination of the energy change of a system during a process involves the evaluation of the energy of the system at the beginning and at the end of the process

$$\text{Energy change} = \text{Energy at final state} - \text{Energy at initial state}$$

$$\Delta E_{system} = E_{final} - E_{initial} = E_2 - E_1$$

- Energy can exist in numerous forms such as internal (sensible, latent, chemical, and nuclear), kinetic, potential, electric, and magnetic
- Their sum constitutes the total energy *E* of a system.

2.6 The First law of Thermodynamics (4)

- In the absence of electric, magnetic, and surface tension effects, the change in the total energy of a system during a process is the sum of the changes in its internal, kinetic, and potential energies

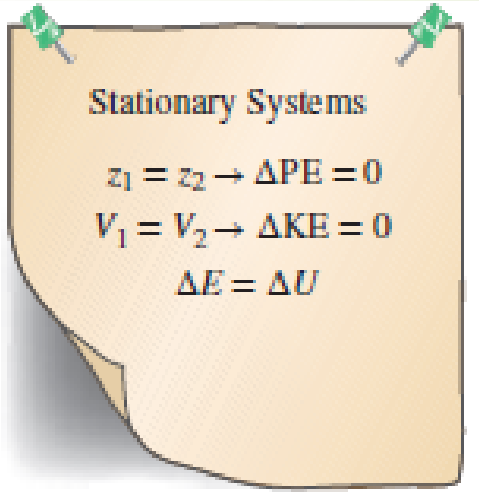
$$\Delta E = \Delta U + \Delta \mathrm{KE} + \Delta \mathrm{PE}$$

where

$$\Delta U = m(u_2 - u_1)$$
$$\Delta \mathrm{KE} = \tfrac{1}{2} m(V_2^2 - V_1^2)$$
$$\Delta \mathrm{PE} = mg(z_2 - z_1)$$

- Most systems encountered in practice are stationary, that is, they do not involve any changes in their velocity or elevation during a process.
- For **stationary systems**,

$$\Delta \mathrm{KE} = \Delta \mathrm{PE} = 0; \text{ thus } \Delta E = \Delta U.$$



2.6 The First law of Thermodynamics (5)

Mechanisms of Energy Transfer, E_{in} and E_{out}

- Energy balance for any system undergoing any kind of process can be expressed more compactly as

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc., energies}}} \qquad \text{(kJ)}$$

or, in the rate form, as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\substack{\text{Rate of net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{dE_{system}/dt}_{\substack{\text{Rate of change in internal,} \\ \text{kinetic, potential, etc., energies}}} \qquad \text{(kW)}$$

- For constant rates, the total quantities during a time interval Δt are related to the quantities per unit time as

$$Q = \dot{Q}\,\Delta t, \quad W = \dot{W}\Delta t, \quad \text{and} \quad \Delta E = (dE/dt)\,\Delta t \qquad \text{(kJ)}$$

2.6 The First law of Thermodynamics (6)

- For a closed system undergoing a **cycle**, the initial and final states are identical, and thus $\Delta E_{system} = E_2 - E_1 = 0$.
- Then the energy balance for a cycle simplifies to $E_{in} - E_{out} = 0$ or $E_{in} = E_{out}$.
- Noting that a closed system does not involve any mass flow across its boundaries, the **energy balance for a cycle** can be expressed in terms of heat and work interactions as

$$W_{net,out} = Q_{net,in} \quad \text{or} \quad \dot{W}_{net,out} = \dot{Q}_{net,in} \qquad \text{(for a cycle)}$$

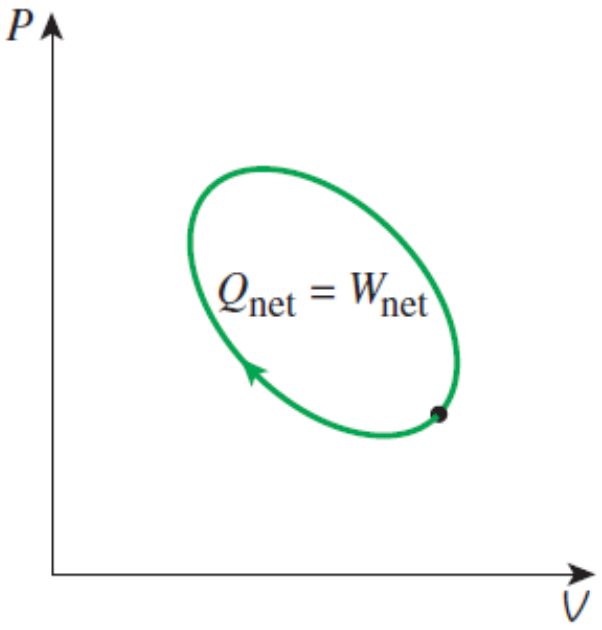


FIGURE 2–48
For a cycle $\Delta E = 0$, thus $Q = W$.

EXAMPLE 2–10 Cooling of a Hot Fluid in a Tank

A rigid tank contains a hot fluid that is cooled while being stirred by a paddle wheel. Initially, the internal energy of the fluid is 800 kJ. During the cooling process, the fluid loses 500 kJ of heat, and the paddle wheel does 100 kJ of work on the fluid. Determine the final internal energy of the fluid. Neglect the energy stored in the paddle wheel.

SOLUTION A fluid in a rigid tank loses heat while being stirred. The final internal energy of the fluid is to be determined.

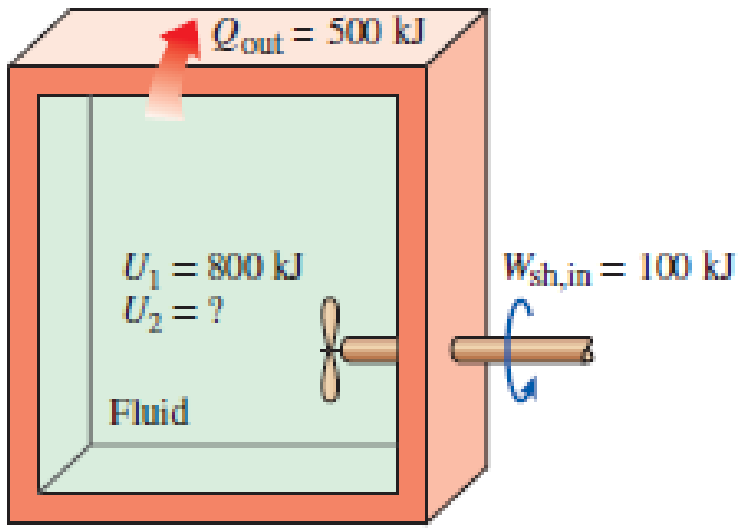


FIGURE 2–49
Schematic for Example 2–10.

Assumptions **1** The tank is stationary and thus the kinetic and potential energy changes are zero, $\Delta KE = \Delta PE = 0$. Therefore, $\Delta E = \Delta U$ and internal energy is the only form of the system's energy that may change during this process. **2** Energy stored in the paddle wheel is negligible.

Analysis Take the contents of the tank as the system (Fig. 2–49). This is a *closed system* since no mass crosses the boundary during the process.

We observe that the volume of a rigid tank is constant, and thus there is no moving boundary work. Also, heat is lost from the system, and shaft work is done on the system. Applying the energy balance on the system gives

$$\underbrace{E_{in} - E_{out}}_{\substack{\text{Net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{system}}_{\substack{\text{Change in internal, kinetic,}\\ \text{potential, etc., energies}}}$$

$$W_{sh,in} - Q_{out} = \Delta U = U_2 - U_1$$

$$100 \text{ kJ} - 500 \text{ kJ} = U_2 - 800 \text{ kJ}$$

$$U_2 = \mathbf{400 \text{ kJ}}$$

Therefore, the final internal energy of the system is 400 kJ.

EXAMPLE 2–11 Acceleration of Air by a Fan

A fan that consumes 20 W of electric power when operating is claimed to discharge air from a ventilated room at a rate of 1.0 kg/s at a discharge velocity of 8 m/s (Fig. 2–50). Determine if this claim is reasonable.

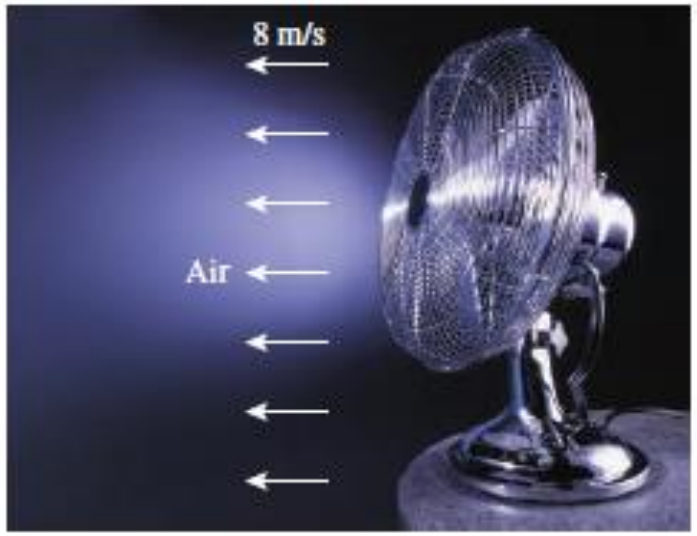


FIGURE 2–50
Schematic for Example 2–11.
©Design Pics/PunchStock RF

SOLUTION A fan is claimed to increase the velocity of air to a specified value while consuming electric power at a specified rate. The validity of this claim is to be investigated.

Assumptions The ventilating room is relatively calm, and air velocity in it is negligible.

Analysis First, let's examine the energy conversions involved: The motor of the fan converts part of the electrical power it consumes to mechanical (shaft) power, which is used to rotate the fan blades in air. The blades are shaped such that they impart a large fraction of the mechanical power of the shaft to air by mobilizing it. In the limiting ideal case of no losses (no conversion of electrical and mechanical energy to thermal energy) in steady operation, the electric power input will be equal to the rate of increase of the kinetic energy of air. Therefore, for a control volume that encloses the fan-motor unit, the energy balance can be written as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\substack{\text{Rate of net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{dE_{system}/dt^{\to 0(\text{steady})}}_{\substack{\text{Rate of change in internal, kinetic,}\\ \text{potential, etc., energies}}} = 0 \to \dot{E}_{in} = \dot{E}_{out}$$

$$\dot{W}_{elect,in} = \dot{m}_{air}\,ke_{out} = \dot{m}_{air}\frac{V_{out}^2}{2}$$

Solving for V_{out} and substituting gives the maximum air outlet velocity to be

$$V_{out} = \sqrt{\frac{2\dot{W}_{elect,in}}{\dot{m}_{air}}} = \sqrt{\frac{2(20\text{ J/s})}{1.0\text{ kg/s}}\left(\frac{1\text{ m}^2/\text{s}^2}{1\text{ J/kg}}\right)} = 6.3\text{ m/s}$$

which is less than 8 m/s. Therefore, the claim is **false**.

Discussion The conservation of energy principle requires the energy to be preserved as it is converted from one form to another, and it does not allow any energy to be created or destroyed during a process. From the first law point of view, there is nothing wrong with the conversion of the entire electrical energy into kinetic energy. Therefore, the first law has no objection to air velocity reaching 6.3 m/s—but this is the upper limit. Any claim of higher velocity is in violation of the first law, and thus impossible. In reality, the air velocity will be considerably lower than 6.3 m/s because of the losses associated with the conversion of electrical energy to mechanical shaft energy, and the conversion of mechanical shaft energy to kinetic energy or air.

EXAMPLE 2–12 Heating Effect of a Fan

A room is initially at the outdoor temperature of 25°C. Now a large fan that consumes 200 W of electricity when running is turned on (Fig. 2–51). The heat transfer rate between the room and the outdoor air is given as $\dot{Q} = UA(T_i - T_o)$ where $U = 6\ \text{W/m}^2\cdot°\text{C}$ is the overall heat transfer coefficient, $A = 30\ \text{m}^2$ is the exposed surface area of the room, and T_i and T_o are the indoor and outdoor air temperatures, respectively. Determine the indoor air temperature when steady operating conditions are established.

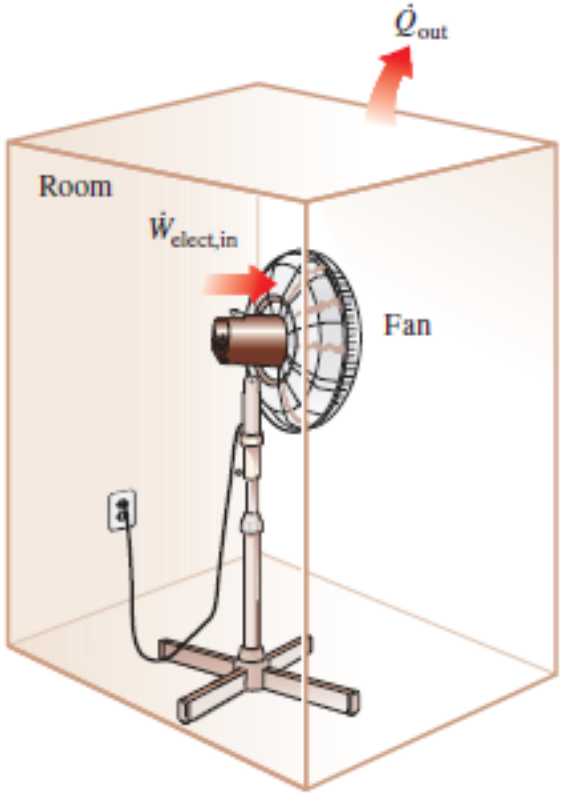


FIGURE 2–51
Schematic for Example 2–12.

SOLUTION A large fan is turned on and kept on in a room that loses heat to the outdoors. The indoor air temperature is to be determined when steady operation is reached.

Assumptions **1** Heat transfer through the floor is negligible. **2** There are no other energy interactions involved.

Analysis The electricity consumed by the fan is energy input for the room, and thus the room gains energy at a rate of 200 W. As a result, the room air temperature tends to rise. But as the room air temperature rises, the rate of heat loss from the room increases until the rate of heat loss equals the electric power consumption. At that point, the temperature of the room air, and thus the energy content of the room, remains constant, and the conservation of energy for the room becomes

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\substack{\text{Rate of net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{dE_{system}/dt^{\nearrow 0(\text{steady})}}_{\substack{\text{Rate of change in internal, kinetic,}\\ \text{potential, etc., energies}}} = 0 \rightarrow \dot{E}_{in} = \dot{E}_{out}$$

$$\dot{W}_{elect,in} = \dot{Q}_{out} = UA(T_i - T_o)$$

Substituting,

$$200\ \text{W} = (6\ \text{W/m}^2\cdot°\text{C})(30\ \text{m}^2)(T_i - 25°\text{C})$$

It gives

$$T_i = \mathbf{26.1°C}$$

Therefore, the room air temperature will remain constant after it reaches 26.1°C.

Discussion Note that a 200-W fan heats a room just like a 200-W resistance heater. In the case of a fan, the motor converts part of the electric energy it draws to mechanical energy in the form of a rotating shaft, while the remaining part is dissipated as heat to the room air because of the motor inefficiency (no motor converts 100 percent of the electric energy it receives to mechanical energy, although some large motors come close with a conversion efficiency of over 97 percent). Part of the mechanical energy of the shaft is converted to kinetic energy of air through the blades, which is then converted to thermal energy as air molecules slow down because of friction. At the end, the entire electric energy drawn by the fan motor is converted to thermal energy of air, which manifests itself as a rise in temperature.

Exercise 1

2–11 Electric power is to be generated by installing a hydraulic turbine–generator at a site 120 m below the free surface of a large water reservoir that can supply water at a rate of 2400 kg/s steadily. Determine the power generation potential.

Exercise 2

2–15 A river flowing steadily at a rate of 175 m^3/s is considered for hydroelectric power generation. It is determined that a dam can be built to collect water and release it from an elevation difference of 80 m to generate power. Determine how much power can be generated from this river water after the dam is filled.

Exercise 3

2–47 Consider a room that is initially at the outdoor temperature of 20°C. The room contains a 40-W lightbulb, a 110-W TV set, a 300-W refrigerator, and a 1200-W iron. Assuming no heat transfer through the walls, determine the rate of increase of the energy content of the room when all of these electric devices are on.

Exercise 4

2–51 An escalator in a shopping center is designed to move 50 people, 75 kg each, at a constant speed of 0.6 m/s at 45° slope. Determine the minimum power input needed to drive this escalator. What would your answer be if the escalator velocity were to be doubled?

2.7 Energy Conversion Efficiencies

- Efficiency is one of the most frequently used terms in thermodynamics, and it indicates how well an energy conversion or transfer process is accomplished.

$$\text{Efficiency} = \frac{\text{Desired output}}{\text{Required input}}$$

- **Efficiency of a water heater**: The ratio of the energy delivered to the house by hot water to the energy supplied to the water heater.

Type	Efficiency
Gas, conventional	55%
Gas, high-efficiency	62%
Electric, conventional	90%
Electric, high-efficiency	94%

FIGURE 2–53
Typical efficiencies of conventional and high-efficiency electric and natural gas water heaters.

Water heater

2.7 Energy Conversion Efficiencies (2)

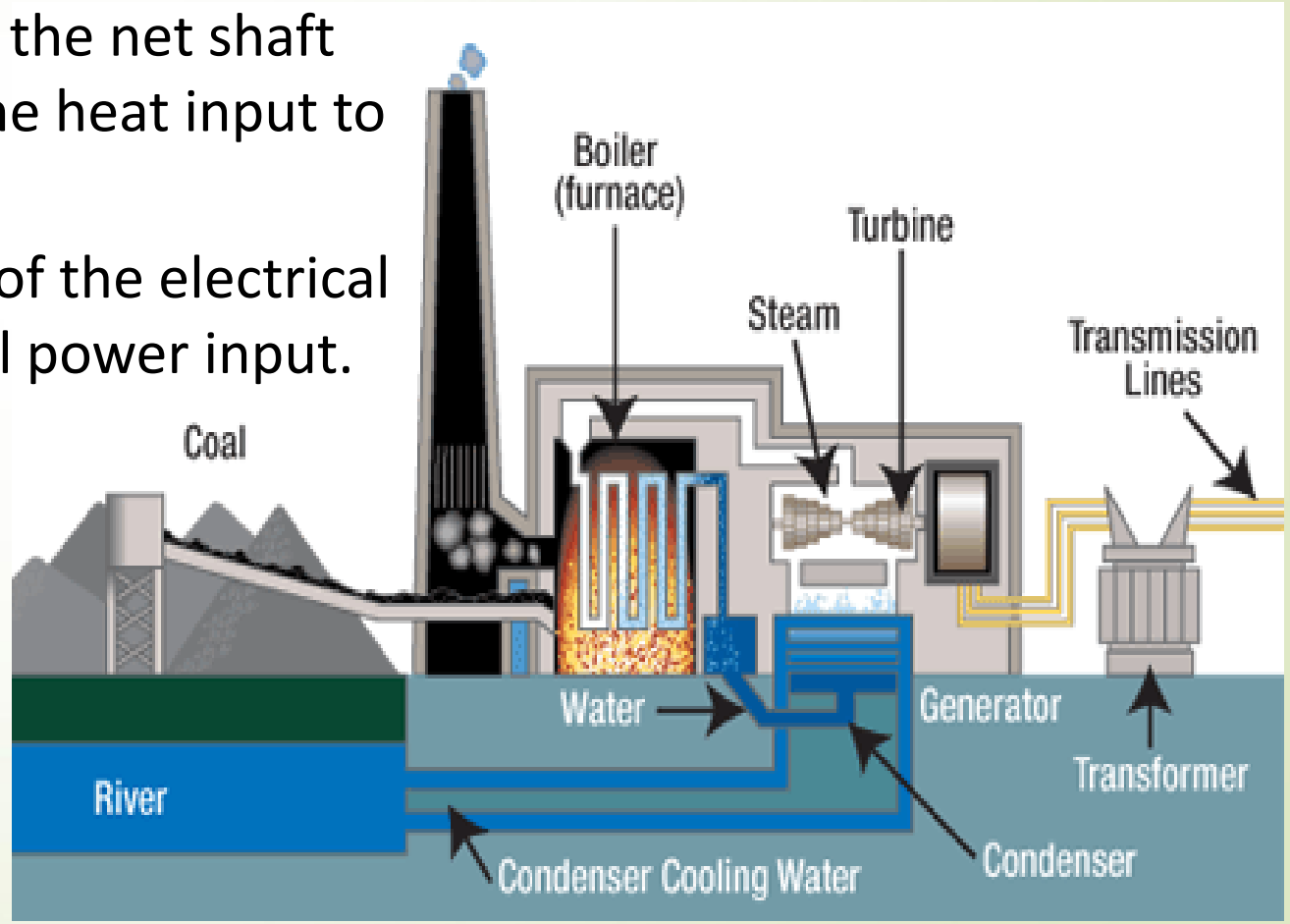
- The efficiency of equipment that involves the combustion of a fuel is based on the **heating value of the fuel:**
  - the amount of heat released when a unit amount of fuel at room temperature is completely burned and the combustion products are cooled to the room temperature.
- Then the performance of combustion equipment can be characterized by **combustion efficiency**, defined as

$$\eta_{\text{combustion}} = \frac{Q}{\text{HV}} = \frac{\text{Amount of heat released during combustion}}{\text{Heating value of the fuel burned}}$$

- Most fuels contain hydrogen, which forms water when burned, and the heating value of a fuel will be different, depending on whether the water in combustion products is in the liquid or vapor form.
  - **Lower heating value (LHV)**: when the water leaves as a vapor, OR
  - **Higher heating value (HHV)**: when the water in the combustion gases is completely condensed and thus the heat of vaporization is also recovered.

2.7 Energy Conversion Efficiencies (3)

- For **power plants**, the work output can be the mechanical power at the turbine exit, or the electrical power output of the generator (a device that converts mechanical energy to electrical energy).
- **Thermal efficiency**: The ratio of the net shaft work output of the turbine to the heat input to the working fluid.
- **Generator efficiency**: The ratio of the electrical power output to the mechanical power input.



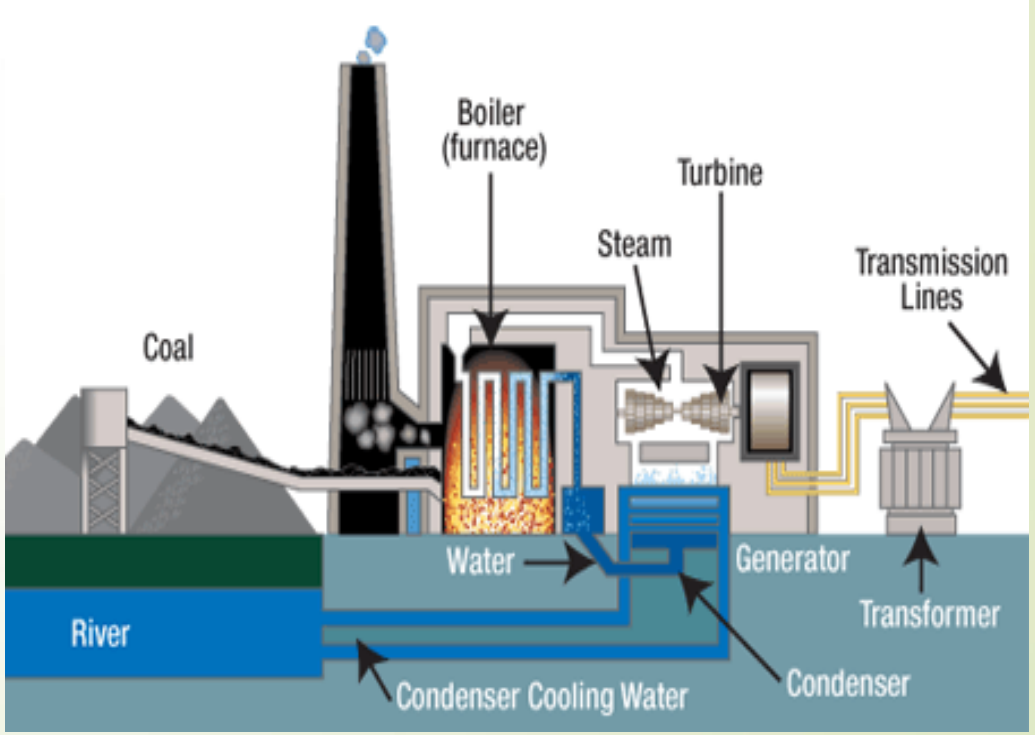
2.7 Energy Conversion Efficiencies (3)

- **Overall efficiency** of a power plant is the ratio of the net electrical power output to the rate of fuel energy input. That is,

$$\eta_{\text{overall}} = \eta_{\text{combustion}}\,\eta_{\text{thermal}}\,\eta_{\text{generator}} = \frac{\dot{W}_{\text{net,electric}}}{\text{HHV} \times \dot{m}_{\text{fuel}}}$$

- **Thermal efficiency**: The ratio of the net shaft work output of the turbine to the heat input to the working fluid.
- **Generator efficiency**: The ratio of the electrical power output to the mechanical power input.

- The overall efficiencies are about
  - Gasoline engine - 25-30%
  - Diesel engines - 35-40%
  - Large power plants – up to 60%



2.7 Energy Conversion Efficiencies (4)

- The efficiency for the **conversion of electricity to light** can be defined as the ratio of the energy converted to light to the electrical energy consumed.
- However, it is more common to express the effectiveness of this conversion process by **lighting efficacy**, which is defined as the amount of light output in lumens per W of electricity consumed.

TABLE 2–1

The efficacy of different lighting systems

Type of lighting	Efficacy, lumens/W
Combustion	
Candle	0.3
Kerosene lamp	1–2
Incandescent	
Ordinary	6–20
Halogen	15–35
Fluorescent	
Compact	40–87
Tube	60–120
High-intensity discharge	
Mercury vapor	40–60
Metal halide	65–118
High-pressure sodium	85–140
Low-pressure sodium	70–200
Solid-State	
LED	20–160
OLED	15–60
Theoretical limit	300*

*This value depends on the spectral distribution of the assumed ideal light source. For white light sources, the upper limit is about 300 lm/W for metal halide, 350 lm/W for fluorescents, and 400 lm/W for LEDs. Spectral maximum occurs at a wavelength of 555 nm (green) with a light output of 683 lm/W.

2.7 Energy Conversion Efficiencies (5)

- The **efficiency of a cooking appliance** can be defined as the ratio of the useful energy transferred to the food to the energy consumed by the appliance.
  - Electric ranges are more efficient than gas ranges, but it is much cheaper to cook with natural gas than with electricity because of the lower unit cost of natural gas.
  - On average, convection ovens save about one-third and microwave ovens save about two-thirds of the energy used by conventional ovens.

TABLE 2–2

Energy costs of cooking a casserole with different appliances*

[From J. T. Amann, A. Wilson, and K. Ackerly, *Consumer Guide to Home Energy Savings*, 9th ed., American Council for an Energy-Efficient Economy, Washington, D.C., 2007, p. 163.]

Cooking appliance	Cooking temperature	Cooking time	Energy used	Cost of energy
Electric oven	350°F (177°C)	1 h	2.0 kWh	$0.19
Convection oven (elect.)	325°F (163°C)	45 min	1.39 kWh	$0.13
Gas oven	350°F (177°C)	1 h	0.112 therm	$0.13
Frying pan	420°F (216°C)	1 h	0.9 kWh	$0.09
Toaster oven	425°F (218°C)	50 min	0.95 kWh	$0.09
Crockpot	200°F (93°C)	7 h	0.7 kWh	$0.07
Microwave oven	"High"	15 min	0.36 kWh	$0.03

EXAMPLE 2–14 Cost of Cooking with Electric and Gas Ranges

The efficiency of cooking appliances affects the internal heat gain from them since an inefficient appliance consumes a greater amount of energy for the same task, and the excess energy consumed shows up as heat in the living space. The efficiency of open burners is determined to be 73 percent for electric units and 38 percent for gas units (Fig. 2–57). Consider a 2-kW electric burner at a location where the unit costs of electricity and natural gas are \$0.12/kWh and \$1.20/therm, respectively. Determine the rate of energy consumption by the burner and the unit cost of utilized energy for both electric and gas burners.

SOLUTION The operation of electric and gas ranges is considered. The rate of energy consumption and the unit cost of utilized energy are to be determined.

Analysis The efficiency of the electric heater is given to be 73 percent. Therefore, a burner that consumes 2 kW of electrical energy will supply

$$\dot{Q}_{\text{utilized}} = (\text{Energy input}) \times (\text{Efficiency}) = (2\ \text{kW})(0.73) = \mathbf{1.46\ kW}$$

of useful energy. The unit cost of utilized energy is inversely proportional to the efficiency and is determined from

$$\text{Cost of utilized energy} = \frac{\text{Cost of energy input}}{\text{Efficiency}} = \frac{\$0.12/\text{kWh}}{0.73} = \mathbf{\$0.164/kWh}$$

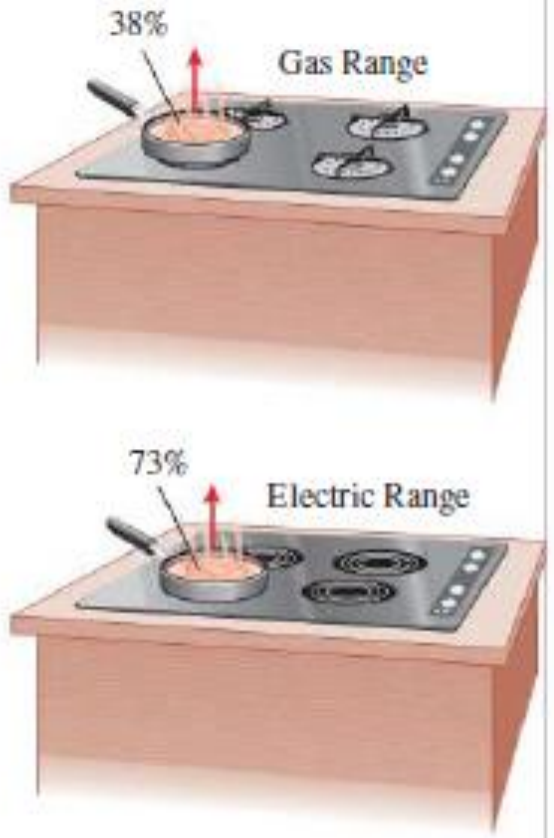


FIGURE 2–57
Schematic of the 73 percent efficient electric heating unit and 38 percent efficient gas burner discussed in Example 2–14.

Noting that the efficiency of a gas burner is 38 percent, the energy input to a gas burner that supplies utilized energy at the same rate (1.46 kW) is

$$\dot{Q}_{\text{input,gas}} = \frac{\dot{Q}_{\text{utilized}}}{\text{Efficiency}} = \frac{1.46\ \text{kW}}{0.38} = \mathbf{3.84\ kW} \qquad (= 13{,}100\ \text{Btu/h})$$

since 1 kW = 3412 Btu/h. Therefore, a gas burner should have a rating of at least 13,100 Btu/h to perform as well as the electric unit.

Noting that 1 therm = 29.3 kWh, the unit cost of utilized energy in the case of a gas burner is determined to be

$$\text{Cost of utilized energy} = \frac{\text{Cost of energy input}}{\text{Efficiency}} = \frac{\$1.20/29.3\ \text{kWh}}{0.38}$$
$$= \mathbf{\$0.108/kWh}$$

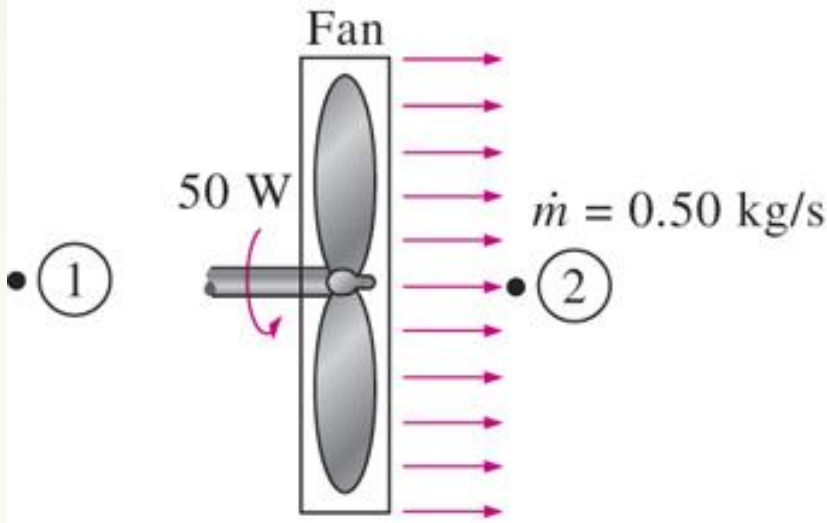
Discussion The cost of utilized gas is less than that of utilized electricity. Therefore, despite its higher efficiency, cooking with an electric burner will cost about 52 percent more compared to a gas burner in this case. This explains why cost-conscious consumers always ask for gas appliances, and it is not wise to use electricity for heating purposes.

2.7 Energy Conversion Efficiencies (6)

Efficiencies of Mechanical and Electrical Devices

- The transfer of mechanical energy is usually accomplished by a rotating shaft, and thus mechanical work is often referred to as shaft work.
  - A **pump** or a **fan** receives shaft work (usually from an electric motor) and transfer it to the fluid as mechanical energy (less frictional losses).
  - A **turbine** on the other hand, converts the mechanical energy of a fluid to shaft work.
- Mechanical efficiency of a device or process can be defined as

$$\eta_{mech} = \frac{Mechanical\ energy\ output}{Mechanical\ energy\ input} = \frac{E_{mech,out}}{E_{mech,in}} = 1 - \frac{E_{mech,loss}}{E_{mech,in}}$$



$V_1 = 0,\ V_2 = 12$ m/s

$z_1 = z_2$

$P_1 = P_2$

$$\eta_{mech,\ fan} = \frac{\Delta\dot{E}_{mech,fluid}}{\dot{W}_{shaft,in}} = \frac{\dot{m}V_2^2/2}{\dot{W}_{shaft,in}} = \frac{(0.50\ \text{kg/s})(12\ \text{m/s})^2/2}{50\ \text{W}} = 0.72$$

2.7 Energy Conversion Efficiencies (7)

- The effectiveness of the conversion process between the mechanical work supplied or extracted and the mechanical energy of the fluid is expressed by the **pump efficiency** and **turbine efficiency**,
- **Pump efficiency**,

$$\eta_{\text{pump}} = \frac{\text{Mechanical energy increase of the fluid}}{\text{Mechanical energy input}} = \frac{\Delta \dot{E}_{\text{mech,fluid}}}{\dot{W}_{\text{shaft,in}}} = \frac{\dot{W}_{\text{pump},u}}{\dot{W}_{\text{pump}}}$$

where $\Delta \dot{E}_{\text{mech,fluid}} = \Delta \dot{E}_{\text{mech,out}} - \Delta \dot{E}_{\text{mech,in}}$ is the rate of increase in the mechanical energy of the fluid, which is equivalent to the **useful pumping power** supplied to the fluid, $\Delta \dot{E}_{\text{mech,fluid}} = \Delta \dot{W}_{\text{pump,u}}$

- **Turbine efficiency**,

$$\eta_{\text{turbine}} = \frac{\text{Mechanical energy output}}{\text{Mechanical energy decrease of the fluid}} = \frac{\dot{W}_{\text{shaft,out}}}{\left|\Delta \dot{E}_{\text{mech,fluid}}\right|} = \frac{\dot{W}_{\text{turbine}}}{\dot{W}_{\text{turbine},e}}$$

where $\left|\Delta \dot{E}_{\text{mech,fluid}}\right| = \Delta \dot{E}_{\text{mech,in}} - \Delta \dot{E}_{\text{mech,out}}$ is the rate of decrease in the mechanical energy of the fluid, which is equivalent to the mechanical **power extracted from the fluid by the turbine**, $\left|\Delta \dot{E}_{\text{mech,fluid}}\right| = \Delta \dot{W}_{\text{turbine,e}}$

2.7 Energy Conversion Efficiencies (8)

- **Motor efficiency,**

$$\eta_{\text{motor}} = \frac{\text{Mechanical power output}}{\text{Electric power input}} = \frac{\dot{W}_{\text{shaft,out}}}{\dot{W}_{\text{elect,in}}}$$

- **Generator efficiency,**

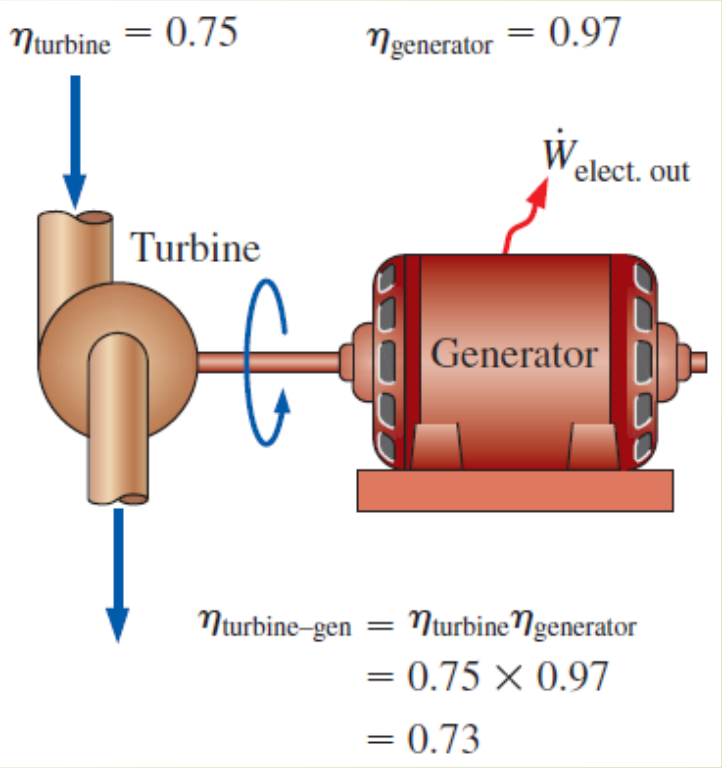
$$\eta_{\text{generator}} = \frac{\text{Electric power output}}{\text{Mechanical power input}} = \frac{\dot{W}_{\text{elect,out}}}{\dot{W}_{\text{shaft,in}}}$$

- **Pump-Motor overall efficiency,**

$$\eta_{\text{pump-motor}} = \eta_{\text{pump}}\eta_{\text{motor}} = \frac{\dot{W}_{\text{pump},u}}{\dot{W}_{\text{elect,in}}} = \frac{\Delta\dot{E}_{\text{mech,fluid}}}{\dot{W}_{\text{elect,in}}}$$

- **Turbine-Generator overall efficiency,**

$$\eta_{\text{turbine-gen}} = \eta_{\text{turbine}}\eta_{\text{generator}} = \frac{\dot{W}_{\text{elect,out}}}{\dot{W}_{\text{turbine},e}} = \frac{\dot{W}_{\text{elect,out}}}{|\Delta\dot{E}_{\text{mech,fluid}}|}$$



EXAMPLE 2–15 Power Generation from a Hydroelectric Plant

Electric power is to be generated by installing a hydraulic turbine–generator at a site 70 m below the free surface of a large water reservoir that can supply water at a rate of 1500 kg/s steadily (Fig. 2–60). If the mechanical power output of the turbine is 800 kW and the electric power generation is 750 kW, determine the turbine efficiency and the combined turbine–generator efficiency of this plant. Neglect losses in the pipes.

SOLUTION A hydraulic turbine–generator installed at a large reservoir is to generate electricity. The combined turbine–generator efficiency and the turbine efficiency are to be determined.

Assumptions **1** The water elevation in the reservoir remains constant. **2** The mechanical energy of water at the turbine exit is negligible.

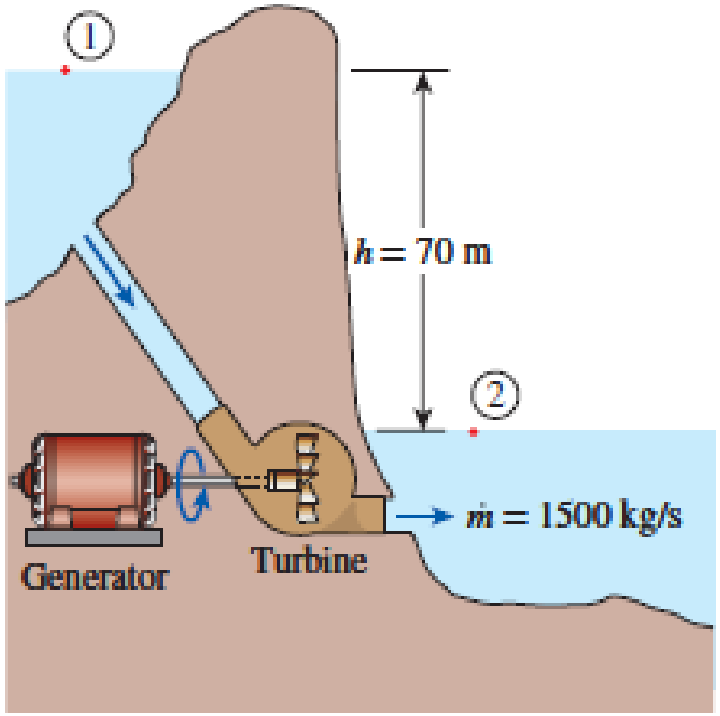


FIGURE 2–60
Schematic for Example 2–15.

Analysis We take the free surface of water in the reservoir to be point 1 and the turbine exit to be point 2. We also take the turbine exit as the reference level ($z_2 = 0$) so that the potential energies at 1 and 2 are $pe_1 = gz_1$ and $pe_2 = 0$. The flow energy P/ρ at both points is zero since both 1 and 2 are open to the atmosphere ($P_1 = P_2 = P_{atm}$). Further, the kinetic energy at both points is zero ($ke_1 = ke_2 = 0$) since the water at point 1 is essentially motionless, and the kinetic energy of water at turbine exit is assumed to be negligible. The potential energy of water at point 1 is

$$pe_1 = gz_1 = (9.81\ \text{m/s}^2)(70\ \text{m})\left(\frac{1\ \text{kJ/kg}}{1000\ \text{m}^2/\text{s}^2}\right) = 0.687\ \text{kJ/kg}$$

Then the rate at which the mechanical energy of water is supplied to the turbine becomes

$$\begin{aligned}\left|\Delta \dot{E}_{\text{mech,fluid}}\right| &= \dot{m}(e_{\text{mech,in}} - e_{\text{mech,out}}) = \dot{m}(\text{pe}_1 - 0) = \dot{m}\text{pe}_1 \\ &= (1500\ \text{kg/s})(0.687\ \text{kJ/kg}) \\ &= 1031\ \text{kW}\end{aligned}$$

The combined turbine–generator and the turbine efficiency are determined from their definitions to be

$$\eta_{\text{turbine-gen}} = \frac{\dot{W}_{\text{elect,out}}}{\left|\Delta \dot{E}_{\text{mech,fluid}}\right|} = \frac{750\ \text{kW}}{1031\ \text{kW}} = 0.727 \text{ or } \mathbf{72.7\%}$$

$$\eta_{\text{turbine}} = \frac{\dot{W}_{\text{shaft,out}}}{\left|\dot{E}_{\text{mech,fluid}}\right|} = \frac{800\ \text{kW}}{1031\ \text{kW}} = 0.776 \text{ or } \mathbf{77.6\%}$$

Therefore, the reservoir supplies 1031 kW of mechanical energy to the turbine, which converts 800 kW of it to shaft work that drives the generator, which then generates 750 kW of electric power.

Discussion This problem can also be solved by taking point 1 to be at the turbine inlet and using flow energy instead of potential energy. It would give the same result since the flow energy at the turbine inlet is equal to the potential energy at the free surface of the reservoir.

EXAMPLE 2–16 Cost Savings Associated with High-Efficiency Motors

A 60-hp electric motor (a motor that delivers 60 hp of shaft power at full load) that has an efficiency of 89.0 percent is worn out and is to be replaced by a 93.2 percent efficient high-efficiency motor (Fig. 2–61). The motor operates 3500 hours a year at full load. Taking the unit cost of electricity to be \$0.08/kWh, determine the amount of energy and money saved as a result of installing the high-efficiency motor instead of the standard motor. Also, determine the simple payback period if the purchase prices of the standard and high-efficiency motors are \$4520 and \$5160, respectively.

SOLUTION A worn-out standard motor is to be replaced by a high-efficiency one. The amount of electrical energy and money saved as well as the simple payback period are to be determined.

Assumptions The load factor of the motor remains constant at 1 (full load) when operating.

Analysis The electric power drawn by each motor and their difference can be expressed as

$$\dot{W}_{\text{electric in, standard}} = \dot{W}_{\text{shaft}}/\eta_{\text{st}} = (\text{Rated power})(\text{Load factor})/\eta_{\text{st}}$$

$$\dot{W}_{\text{electric in, efficient}} = \dot{W}_{\text{shaft}}/\eta_{\text{eff}} = (\text{Rated power})(\text{Load factor})/\eta_{\text{eff}}$$

$$\begin{aligned}\text{Power savings} &= \dot{W}_{\text{electric in, standard}} - \dot{W}_{\text{electric in, efficient}} \\ &= (\text{Rated power})(\text{Load factor})(1/\eta_{\text{st}} - 1/\eta_{\text{eff}})\end{aligned}$$

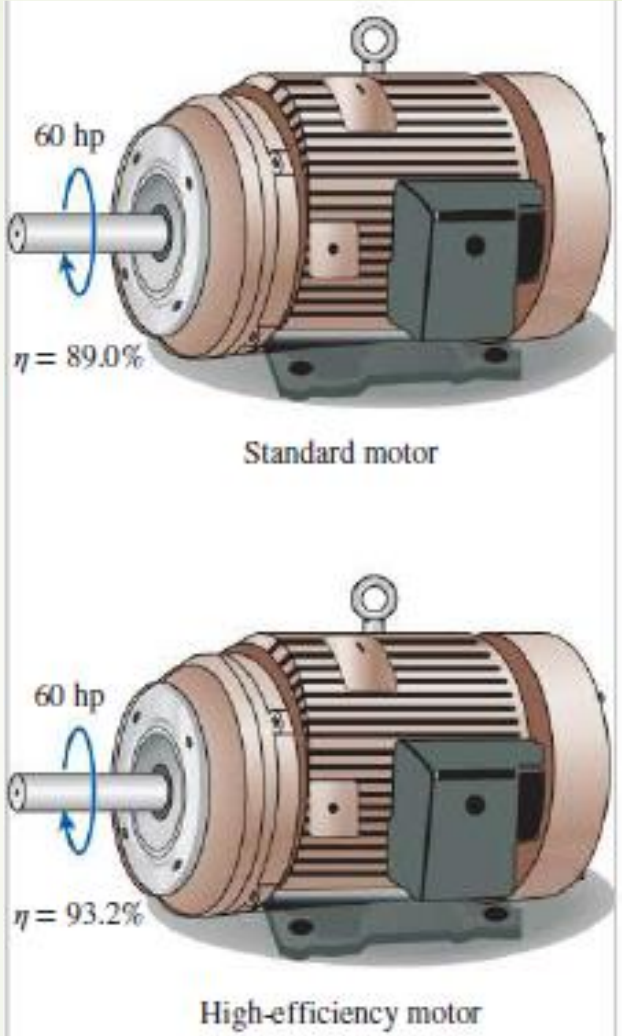


FIGURE 2–61
Schematic for Example 2–16.

where η_{st} is the efficiency of the standard motor, and η_{eff} is the efficiency of the comparable high-efficiency motor. Then the annual energy and cost savings associated with the installation of the high-efficiency motor become

$$\begin{aligned}\text{Energy savings} &= (\text{Power savings})(\text{Operating hours})\\ &= (\text{Rated power})(\text{Operating hours})(\text{Load factor})(1/\eta_{st} - 1\eta_{eff})\\ &= (60\text{ hp})(0.7457\text{ kW/hp})(3500\text{ h/year})(1)(1/0.89 - 1/0.932)\\ &= \mathbf{7929\text{ kWh/year}}\end{aligned}$$

$$\begin{aligned}\text{Cost savings} &= (\text{Energy savings})(\text{Unit cost of energy})\\ &= (7929\text{ kWh/year})(\$0.08/\text{kWh})\\ &= \mathbf{\$634/\text{year}}\end{aligned}$$

Also,

$$\text{Excess initial cost} = \text{Purchase price differential} = \$5160 - \$4520 = \$640$$

This gives a simple payback period of

$$\text{Simple payback period} = \frac{\text{Excess initial cost}}{\text{Annual cost savings}} = \frac{\$640}{\$634/\text{year}} = \mathbf{1.01\text{ year}}$$

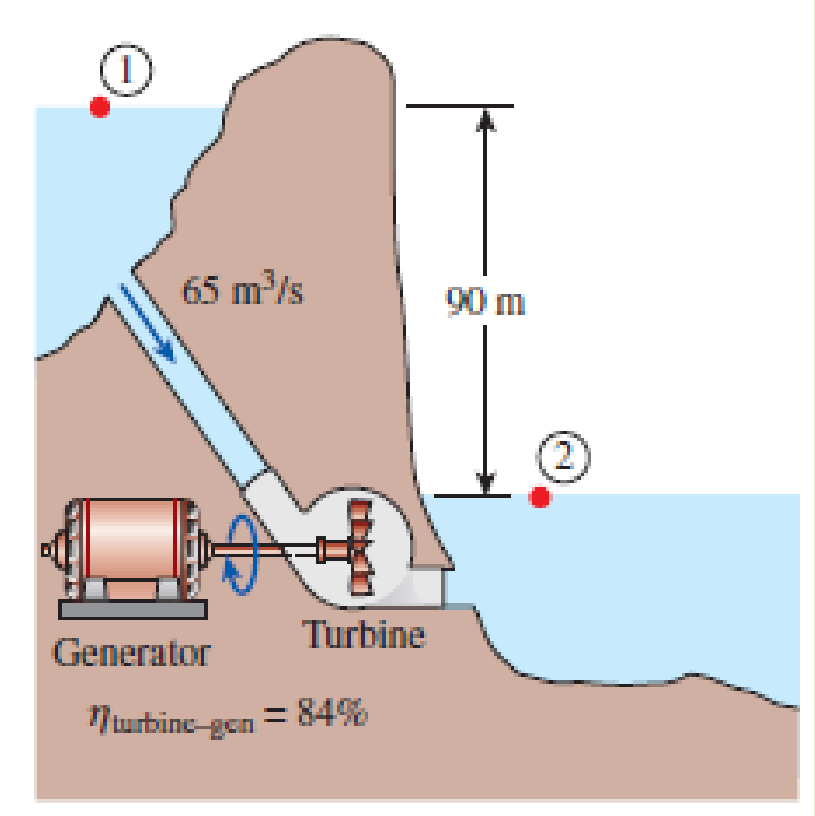
Discussion Note that the high-efficiency motor pays for its price differential within about one year from the electrical energy it saves. Considering that the service life of electric motors is several years, the purchase of the higher efficiency motor is definitely indicated in this case.

Exercise 5

2–64 The water in a large lake is to be used to generate electricity by the installation of a hydraulic turbine-generator at a location where the depth of the water is 50 m. Water is to be supplied at a rate of 5000 kg/s. If the electric power generated is measured to be 1862 kW and the generator efficiency is 95 percent, determine (*a*) the overall efficiency of the turbine—generator, (*b*) the mechanical efficiency of the turbine, and (*c*) the shaft power supplied by he turbine to the generator.

Exercise 6

2–122 In a hydroelectric power plant, 65 m^3/s of water flows from an elevation of 90 m to a turbine, where electric power is generated. The overall efficiency of the turbine–generator is 84 percent. Disregarding frictional losses in piping, estimate the electric power output of this plant. *Answer:* 48.2 MW



2.8 Energy and Environment

- The conversion of energy from one form to another often affects the environment and the air we breathe in many ways, and thus the study of energy is not complete without considering its impact on the environment.
- Pollutants emitted during the combustion of fossil fuels are responsible for smog, acid rain, and global warming.
- The environmental pollution has reached such high levels that it became a serious threat to vegetation, wildlife, and human health.

FIGURE 2–63
Motor vehicles are the largest source of air pollution.

FIGURE 2–62
Energy conversion processes are often accompanied by environmental pollution.

2.8 Energy and Environment (2)

Ozone and Smog

- **Smog**: Made up mostly of ground-level ozone (O3), but it also contains numerous other chemicals, including carbon monoxide (CO), particulate matter such as soot and dust, volatile organic compounds (VOCs) such as benzene, butane, and other hydrocarbons.
- **Hydrocarbons** and **nitrogen oxides** react in the presence of sunlight on hot calm days to form ground-level ozone.
- **Ozone** irritates eyes and damages the air sacs in the lungs where oxygen and carbon dioxide are exchanged, causing eventual hardening of this soft and spongy tissue.
- It also causes shortness of breath, wheezing, fatigue, headaches, and nausea, and aggravates respiratory problems such as asthma.

2.8 Energy and Environment (3)

- The other serious pollutant in smog is **carbon monoxide**, which is a colorless, odorless, poisonous gas. It is mostly emitted by motor vehicles.
- It deprives the body's organs from getting enough oxygen by binding with the red blood cells that would otherwise carry oxygen. It is fatal at high levels.
- Suspended **particulate matter** such as dust and soot are emitted by vehicles and industrial facilities. Such particles irritate the eyes and the lungs.

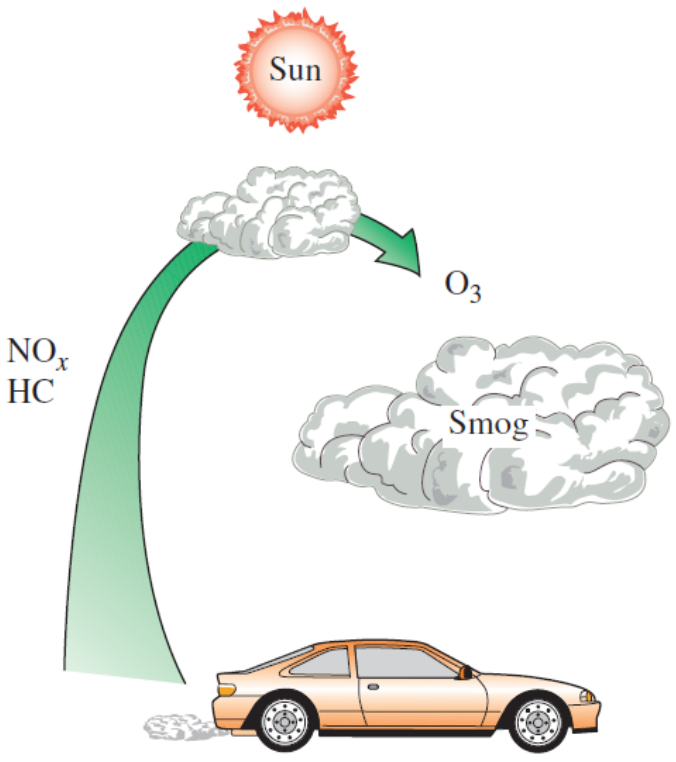


FIGURE 2–64
Ground-level ozone, which is the primary component of smog, forms when HC and NO_x react in the presence of sunlight in hot calm days.

2.8 Energy and Environment (4)

Acid Rain

- The sulfur in the fuel reacts with oxygen to form sulfur dioxide (SO2), which is an air pollutant.
- The main source of SO2 is the electric power plants that burn high-sulfur coal.
- Motor vehicles also contribute to SO2 emissions since gasoline and diesel fuel also contain small amounts of sulfur.
- The sulfur oxides and nitric oxides react with water vapor and other chemicals high in the atmosphere in the presence of sunlight to form sulfuric and nitric acids.

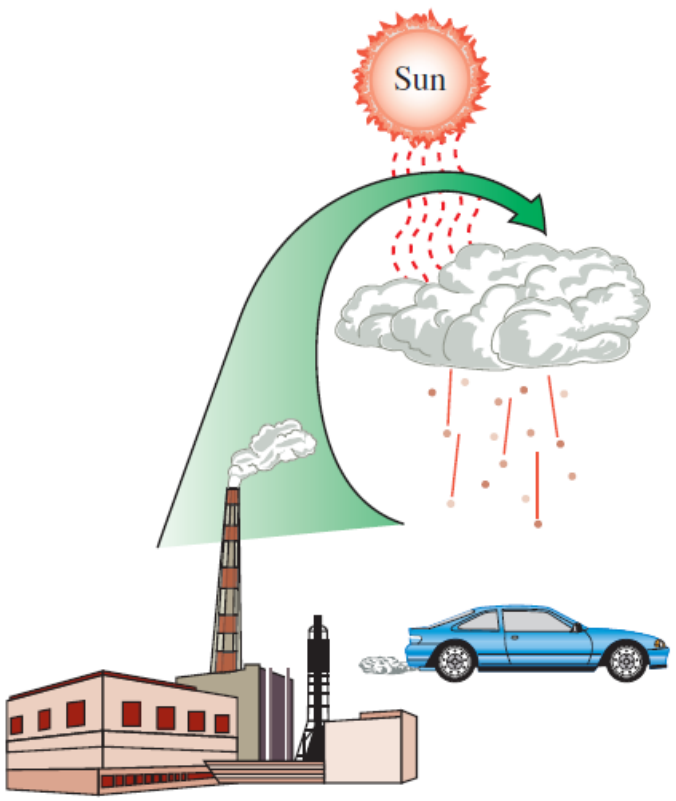


FIGURE 2–65
Sulfuric acid and nitric acid are formed when sulfur oxides and nitric oxides react with water vapor and other chemicals high in the atmosphere in the presence of sunlight.

- The acids formed usually dissolve in the suspended water droplets in clouds or fog.
- These acid-laden droplets, which can be as acidic as lemon juice, are washed from the air on to the soil by rain or snow. This is known as **acid rain**.

2.8 Energy and Environment (5)

The Greenhouse Effect: Global Warming and Climate Change

- **Greenhouse effect**:
  - Glass allows the solar radiation to enter freely but blocks the infrared radiation emitted by the interior surfaces.
  - This causes a rise in the interior temperature as a result of the thermal energy buildup in a space.
- The surface of the earth, which warms up during the day as a result of the absorption of solar energy, cools down at night by radiating part of its energy into deep space as infrared radiation.

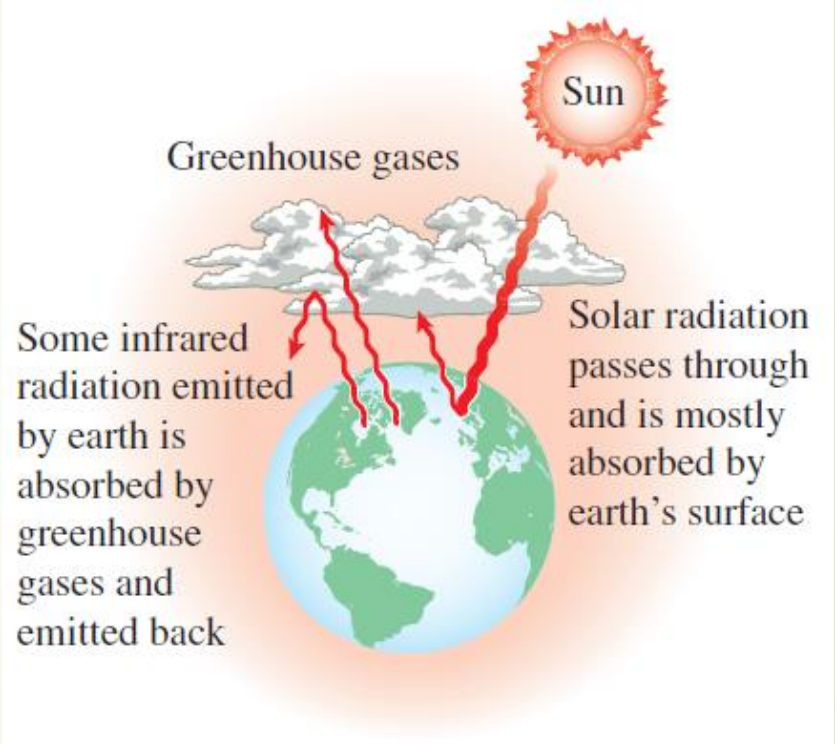


FIGURE 2–66
The greenhouse effect on earth.

2.8 Energy and Environment (6)

- **Carbon dioxide (CO_2)**, water vapor, and trace amounts of some other gases such as methane and nitrogen oxides act like a blanket and keep the earth warm at night by blocking the heat radiated from the earth.
- The result is **global warming**.
- These gases are called "**greenhouse gases**," with CO_2 being the primary component.
  - Under normal conditions, CO_2 amount is controlled and balanced by vegetation through the process of photosynthesis, which releases O_2.
  - However, deforestation and excessive CO_2 production disturb the balance.
- The greenhouse effect makes life on earth possible by keeping the earth warm. However excessive amounts of these gasses trapping too much energy causing global warming.

FIGURE 2–67

The average car produces several times its weight in CO_2 every year (it is driven 20,000 km a year, consumes 2300 liters of gasoline, and produces 2.5 kg of CO_2 per liter).

FIGURE 2–68

Renewable energies such as wind are called "green energy" since they emit no pollutants or greenhouse gases.

End